🎯 2026 PRELIMS & RO/ARO SPECIAL · COMPLETE CHAPTER-WISE PYQS

SCIENCE & TECHNOLOGY PYQ MASTER

सामान्य विज्ञान एवं प्रौद्योगिकी — अध्यायवार संपूर्ण प्रश्नोत्तरी

Consolidated Chapter-wise Practice Bank covering all 26 topics.

Exclusive Authentic UPPCS & UP RO/ARO Questions Only • Structured Option Grids • Visible Answers & High-Yield Core Memory Lock Explanations.

UPPCS & UP RO/ARO Prelims 2026 • Active Recall Question Bank Series • 100% Exam-Verified Edition

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Chapter 01: Living World, Cell, Genetics and Biotechnology — UPPCS & UP RO/ARO Drill

15 Authentic Questions • Direct Verified Answers • Active Recall Memory Locks

🎯 High-Yield PYQ Drill (UPPCS & UP RO/ARO)

Exam Drill Rule: High-yield authentic UPPCS & UP RO/ARO questions. Attempt actively; verify against the prominent Memory Lock box below each question.

Q1 UPPCS PRELIMS 2025

With reference to Whittaker's five-kingdom classification, which of the following statements is/are correct?
1. Monera includes bacteria, which are prokaryotes with peptidoglycan walls.
2. Viruses form the sixth kingdom in this phylogenetic system.
3. Fungi are placed in Plantae because both groups possess rigid cell walls.

Select the correct answer from the code given below:

- **A** 1 and 3
- **B** Only 1
- **C** 2 and 3
- **D** Only 2

Correct Answer: (B)
Core Memory Fact: Logic:** Standard key matches Only 1. **Ans: B.** Only 1

Q2 UPPCS PRELIMS 2024

Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).
Assertion (A): In a double-stranded DNA molecule, the ratio of $(A+G)$ to $(T+C)$ is always equal to 1.
Reason (R): A purine base always pairs strictly with a pyrimidine base via hydrogen bonding across the two antiparallel strands.

Select the correct answer from the code given below:

- **A** Both (A) and (R) are true, but (R) is not the correct explanation of (A)
- **B** (A) is false, but (R) is true
- **C** (A) is true, but (R) is false
- **D** Both (A) and (R) are true and (R) is the correct explanation of (A)

Correct Answer: (D)
Core Memory Fact: Logic:** According to Chargaff's rules, in double-stranded DNA, Adenine pairs with Thymine ($A = T$) and Guanine pairs with Cytosine ($G = C$). **Ans: D.** Both (A) and (R) are true and (R) is the correct explanation of (A)

Q3 UP RO/ARO PRELIMS 2023

Match List-I with List-II:

List-I (Scientist)	List-II (Discovery / Concept)
A Robert Hooke	1 Identified ribosomes
B Robert Brown	2 Postulated *Omnis cellula-e cellula*
C George Palade	3 Discovered the cell nucleus
D Rudolf Virchow	4 Coined the word cell from cork

Select the correct answer from the code given below:

A. A-4, B-3, C-2, D-1
B. A-1, B-2, C-3, D-4
C. A-4, B-3, C-1, D-2
D. A-2, B-4, C-1, D-3

Correct Answer: (C)

Core Memory Fact: Logic:** Robert Hooke (1665) observed honey-comb cork compartments and coined "cell" (A-4). **Ans: C.** A-4, B-3, C-1, D-2

Q4 UPPCS PRELIMS 2023

With reference to cell division, which of the following statements is/are correct?
1. Meiosis results in four genetically diverse haploid daughter nuclei.
2. Crossing over occurs during the pachytene stage of prophase-I in meiosis.
3. DNA replication occurs during the G_2 phase of interphase.

Select the correct answer from the code given below:

A. 1 and 2 only
B. 2 and 3 only
C. 1, 2 and 3
D. Only 1

Correct Answer: (A)

Core Memory Fact: Logic:** Statements 1 and 2 are correct: Meiosis is a reductional division producing 4 haploid gametes; crossing over between non-sister chromatids occurs at pachytene mediated by recombinase. **Ans: A.** 1 and 2 only

Q5 UPPCS PRELIMS 2022

Which of the following pairs is **not** correctly matched?

A. Kary Mullis — Polymerase Chain Reaction (PCR)
B. Hugo de Vries — Laws of inheritance worked out on garden pea
C. Alec Jeffreys — DNA Fingerprinting
D. Ian Wilmut — Dolly the cloned sheep

Correct Answer: (B)

Core Memory Fact: Logic:** The laws of inheritance using the garden pea were discovered by Gregor Johann Mendel. **Ans: B.** Hugo de Vries — Laws of inheritance worked out on garden pea

Q6 UP RO/ARO PRELIMS 2021

Arrange the following biological milestones in correct chronological order:
1. Watson and Crick's double-helix model of DNA
2. Robert Hooke's description of cork cells
3. Rudolf Virchow's doctrine *Omnis cellula-e cellula*
4. Schleiden and Schwann's formulation of Cell Theory

Select the correct answer from the code given below:

A 1, 2, 4, 3
B 2, 3, 4, 1
C 4, 2, 3, 1
D 2, 4, 3, 1

Correct Answer: (D)
Core Memory Fact: Logic:** Standard key matches 2, 4, 3, 1. **Ans: D.** 2, 4, 3, 1

Q7 UPPCS PRELIMS 2021

With reference to nucleic acids and genetic codons, which of the following statements is/are correct?
1. AUG functions as the universal start codon and codes for Methionine.
2. UAA, UAG, and UGA are nonsense termination codons.
3. RNA contains Thymine in place of Uracil.
4. Human somatic diploid cells contain 46 chromosomes.

Select the correct answer from the code given below:

A 1 and 3
B Only 4
C 1, 2 and 4
D 2 and 3

Correct Answer: (C)
Core Memory Fact: Logic:** Statements 1, 2, and 4 are correct. **Ans: C.** 1, 2 and 4

Q8 UPPCS PRELIMS 2020

Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).
Assertion (A): *Agrobacterium tumefaciens* is widely referred to as the natural genetic engineer of plants.
Reason (R): It naturally carries a Ti plasmid that transfers a specific segment of DNA (T-DNA) into the genome of host plant cells.

Select the correct answer from the code given below:

A Both (A) and (R) are true, but (R) is not the correct explanation of (A)
B (A) is false, but (R) is true
C (A) is true, but (R) is false
D Both (A) and (R) are true and (R) is the correct explanation of (A)

Correct Answer: (D)
Core Memory Fact: Logic:** Agrobacterium tumefaciens is called the natural genetic engineer precisely because its Ti plasmid integrates T-DNA directly into plant chromosomes, making it an ideal modified vector for transgenic plant creation. **Ans: D.** Both (A) and (R) are true and (R) is the correct explanation of (A)

Q9 UPPCS (MAINS) GS-1

With reference to prokaryotic bacterial cells, which of the following statements is/are correct?
1. They lack membrane-bound cellular organelles such as mitochondria and chloroplasts.
2. Their cytoplasmic ribosomes are 70S.
3. Their cell walls contain peptidoglycan.

Select the correct answer from the code given below:

- **A** 1 and 2 only
- **B** 2 and 3 only
- **C** 1 and 3 only
- **D** 1, 2 and 3

Correct Answer: (D)
Core Memory Fact: Logic:** All three statements accurately describe prokaryotic bacteria: they lack membrane-enclosed organelles; have 70S ribosomes (50S + 30S); and maintain cell shape via peptidoglycan (murein) walls. **Ans: D.** 1, 2 and 3

Q10 UPPCS PRELIMS 2019

Match List-I with List-II:

List-I (Biotechnology Product / Tool)	List-II (Key Association / Feature)
A Humulin	**1** Transgenic pest-resistant cotton
B Ti Plasmid	**2** Recombinant human insulin (1982)
C *Bacillus thuringiensis*	**3** Gene vector from *Agrobacterium*
D *Thermus aquaticus*	**4** Source of heat-stable *Taq* polymerase

Select the correct answer from the code given below:

- **A** A-2, B-3, C-1, D-4
- **B** A-3, B-2, C-1, D-4
- **C** A-2, B-3, C-4, D-1
- **D** A-1, B-4, C-2, D-3

Correct Answer: (A)
Core Memory Fact: Logic:** Humulin = First recombinant human insulin produced by Eli Lilly (A-2). **Ans: A.** A-2, B-3, C-1, D-4

Q11 UP RO/ARO 2016

In the human cell, which of the following organelles contains its own genetic material (DNA) and 70S ribosomes?

- **A** Endoplasmic reticulum
- **B** Mitochondria
- **C** Golgi apparatus
- **D** Lysosome

Correct Answer: (B)
Core Memory Fact: Logic:** Mitochondria and chloroplasts are semi-autonomous organelles containing their own circular double-stranded DNA and 70S ribosomes, reflecting their endosymbiotic prokaryotic ancestry. **Ans: B.** Mitochondria

Q12 UPPCS PRELIMS 2018

In meiosis, the physical exchange of genetic material (crossing over) between non-sister chromatids occurs during which specific stage?

A Leptotene
B Zygotene
C Pachytene
D Diakinesis

Correct Answer: (C)
Core Memory Fact: Logic:** Crossing over takes place during the pachytene stage of prophase-I, mediated by the enzyme recombinase. **Ans: C.** Pachytene

Q13 UPPCS (PRE) 2017

Which of the following human genetic disorders is caused by an extra copy of chromosome 21 (Trisomy 21)?

A Klinefelter syndrome
B Turner syndrome
C Down syndrome
D Thalassemia

Correct Answer: (C)
Core Memory Fact: Logic:** Down syndrome is caused by trisomy of autosome 21 ($47, 21+1$). **Ans: C.** Down syndrome

Q14 UPPCS (PRE) 2016

Sickle-cell anaemia is caused by a point mutation in the β-globin gene that results in the replacement of:

A Valine by Glutamic acid
B Glutamic acid by Valine
C Alanine by Glycine
D Leucine by Proline

Correct Answer: (B)
Core Memory Fact: Logic:** In sickle-cell anaemia, a transversion mutation ($GAG \rightarrow GUG$) substitutes hydrophilic Glutamic acid with hydrophobic Valine at position 6 of the β-globin polypeptide chain. **Ans: B.** Glutamic acid by Valine

Q15 UP RO/ARO 2014

The chemical substance known as Colchicine, used in plant breeding to induce polyploidy, acts by:

A Inhibiting DNA replication in S-phase
B Preventing mitotic spindle formation, arresting cells at metaphase
C Accelerating cytokinesis
D Promoting rapid crossing over

Correct Answer: (B)
Core Memory Fact: Logic:** Colchicine inhibits microtubule polymerisation, preventing mitotic spindle assembly and arresting dividing cells at metaphase, causing chromosome doubling (polyploidy). **Ans: B.** Preventing mitotic spindle formation, arresting cells at metaphase

Chapter 02: Animal Biology and Husbandry — UPPCS & UP RO/ARO Drill

15 Authentic Questions • Direct Verified Answers • Active Recall Memory Locks

🎯 High-Yield PYQ Drill (UPPCS & UP RO/ARO)

Exam Drill Rule: High-yield authentic UPPCS & UP RO/ARO questions. Attempt actively; verify against the prominent Memory Lock box below each question.

Q1 UPPCS PRELIMS 2025

With reference to respiration in animals, which of the following statements is/are correct?
1. Spiders and scorpions breathe primarily through book lungs.
2. Insects transport oxygen throughout their body primarily via haemocyanin dissolved in haemolymph.
3. Whales and dolphins possess branchial gills during their embryonic stages only.

Select the correct answer from the code given below:

- **A** Only 1
- **B** 1 and 2
- **C** 2 and 3
- **D** 1, 2 and 3

Correct Answer: (A)

Core Memory Fact: Logic:** Standard key matches Only 1. **Ans: A.** Only 1

Q2 UPPCS PRELIMS 2024

Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).
Assertion (A): Natural silk burns with the characteristic smell of burning hair, whereas pure cotton burns with the smell of burning paper.
Reason (R): Natural silk is an animal protein composed of fibroin, whereas cotton is a plant polysaccharide composed of cellulose.

Select the correct answer from the code given below:

- **A** Both (A) and (R) are true, but (R) is not the correct explanation of (A)
- **B** (A) is false, but (R) is true
- **C** (A) is true, but (R) is false
- **D** Both (A) and (R) are true and (R) is the correct explanation of (A)

Correct Answer: (D)

Core Memory Fact: Logic:** Silk and hair are both proteins containing nitrogenous peptide bonds that emit the smell of burning keratin/fibroin when burned. **Ans: D.** Both (A) and (R) are true and (R) is the correct explanation of (A)

Q3 UP RO/ARO PRELIMS 2023

Match List-I with List-II:

List-I (Animal Product / Breed)	List-II (Source Organism / Origin)
A Pashmina	1 *Pinctada vulgaris* (Mollusc)
B Muga Silk	2 Changthangi Goat of Ladakh
C Pearl	3 *Antheraea assamensis* of Assam
D Black Gold Buffalo	4 Murrah breed of Haryana

Select the correct answer from the code given below:

A A-2, B-3, C-1, D-4

B A-3, B-2, C-1, D-4

C A-2, B-1, C-3, D-4

D A-4, B-3, C-1, D-2

Correct Answer: (A)

Core Memory Fact: Logic:** Pashmina = Fine undercoat of Changthangi goat (A-2). **Ans: A.** A-2, B-3, C-1, D-4

Q4 UPPCS PRELIMS 2023

Which of the following animal groups has a four-chambered heart?

A Shark, Frog, Lizard

B Crocodile, Pigeon, Whale

C Rohu, Chameleon, Ostrich

D Toad, Cobra, Bat

Correct Answer: (B)

Core Memory Fact: Logic:** All birds (pigeon) and mammals (whale) have a 4-chambered heart. **Ans: B.** Crocodile, Pigeon, Whale

Q5 UPPCS PRELIMS 2022

Which of the following buffalo breeds is indigenous to Uttar Pradesh and is renowned for the highest milk fat percentage (up to 13%)?

A Murrah

B Jaffarabadi

C Bhadawari

D Nili-Ravi

Correct Answer: (C)

Core Memory Fact: Logic:** The Bhadawari buffalo, native to the Agra–Etawah belt of UP and Bhind–Morena in MP, yields milk with the highest fat percentage (up to 12.5–13%). **Ans: C.** Bhadawari

Q6 UP RO/ARO PRELIMS 2021

The water vascular (ambulacral) system operating tube feet is the unique distinguishing characteristic of which animal phylum?

A Mollusca
B Arthropoda
C Echinodermata
D Annelida

Correct Answer: (C)
Core Memory Fact: Logic:** Echinoderms (starfish, sea urchins, sea cucumbers) uniquely possess a hydraulic water vascular system for locomotion and prey capture. **Ans: C.** Echinodermata

Q7 UPPCS PRELIMS 2021

Which of the following animals is a true pseudoruminant, possessing an active fermentation rumen but lacking the omasum?

A Cow
B Camel
C Goat
D Buffalo

Correct Answer: (B)
Core Memory Fact: Logic:** Standard key matches Camel. **Ans: B.** Camel

Q8 UPPCS PRELIMS 2020

Flame cells (solenocytes) are the specialized excretory and osmoregulatory structures of:

A Roundworms (Nematoda)
B Segmented worms (Annelida)
C Flatworms (Platyhelminthes)
D Sponges (Porifera)

Correct Answer: (C)
Core Memory Fact: Logic:** Flame cells are found in flatworms (Planaria, Taenia, Fasciola). **Ans: C.** Flatworms / Platyhelminthes

Q9 UPPCS (MAINS) GS-1

With reference to livestock revolutions in India, which of the following pairs is **not correctly matched?**

A White Revolution — Milk production (Dr. Verghese Kurien)
B Silver Revolution — Egg and poultry production
C Blue Revolution — Fish production
D Yellow Revolution — Cattle meat production

Correct Answer: (D)
Core Memory Fact: Logic:** Yellow Revolution refers to Oilseeds production (mustard, sunflower). **Ans: D.** Yellow Revolution — Cattle meat production

Q10 UPPCS PRELIMS 2019

The Kadaknath fowl, renowned for its black-coloured meat, bones, and internal organs, is native to which Indian state?

- **A** Uttar Pradesh
- **B** Madhya Pradesh
- **C** Rajasthan
- **D** Andhra Pradesh

Correct Answer: (B)

Core Memory Fact: Logic:** Kadaknath (Kalamasi) is native to Jhabua and Dhar districts of Madhya Pradesh and holds a GI tag. **Ans: B.** Madhya Pradesh

Q11 UP RO/ARO 2016

Which of the following is an egg-laying mammal (Monotreme)?

- **A** Kangaroo
- **B** Bat
- **C** Duck-billed Platypus
- **D** Blue Whale

Correct Answer: (C)

Core Memory Fact: Logic:** The Duck-billed Platypus (Ornithorhynchus) and Echidna are the world's only oviparous (egg-laying) mammals. **Ans: C.** Duck-billed Platypus

Q12 UPPCS PRELIMS 2018

In honeybee colonies, the famous "waggle dance" performed by scout bees communicates:

- **A** Threat of predatory hornets
- **B** Presence of queen bee pheromone
- **C** Distance and direction of distant floral nectar sources
- **D** Swarming time

Correct Answer: (C)

Core Memory Fact: Logic:** Karl von Frisch demonstrated that the waggle dance communicates distance ($>50\text{ m}$) and angle relative to the sun to locate rich nectar flowers. **Ans: C.** Distance and direction of distant floral nectar sources

Q13 UPPCS (PRE) 2017

Which of the following silkworms feeds on the leaves of Arjun (*Terminalia arjuna*) and Asan trees?

- **A** Mulberry silkworm (*Bombyx mori*)
- **B** Tasar silkworm (*Antheraea mylitta*)
- **C** Muga silkworm (*Antheraea assamensis*)
- **D** Eri silkworm (*Samia ricini*)

Correct Answer: (B)

Core Memory Fact: Logic:** Tasar silkworms feed on Arjun and Asan trees in Jharkhand and Chhattisgarh. **Ans: B.** Tasar silkworm

Q14 UPPCS (PRE) 2016

Foot and Mouth Disease (FMD) affecting cloven-hoofed domestic livestock is caused by a:

A Bacterium
B Fungus
C Virus
D Protozoan

Correct Answer: (C)
Core Memory Fact: Logic:** Standard key matches Virus. **Ans: C.** Virus

Q15 UP RO/ARO 2014

The Ganges River Dolphin (*Platanista gangetica*), India's National Aquatic Animal, navigates and hunts prey using:

A Keen underwater vision
B Electro-reception via ampullae
C High-frequency ultrasonic echolocation
D Olfactory scent trails

Correct Answer: (C)
Core Memory Fact: Logic:** The Ganges river dolphin has a rudimentary lensless eye (functionally blind) and relies strictly on ultrasonic bio-sonar / echolocation to navigate muddy river waters. **Ans: C.** High-frequency ultrasonic echolocation

Chapter 03: Nutrition, Vitamins, Diseases and Medicine — UPPCS & UP RO/ARO Drill

15 Authentic Questions • Direct Verified Answers • Active Recall Memory Locks

🎯 High-Yield PYQ Drill (UPPCS & UP RO/ARO)

Exam Drill Rule: High-yield authentic UPPCS & UP RO/ARO questions. Attempt actively; verify against the prominent Memory Lock box below each question.

Q1 UPPCS PRELIMS 2025

With reference to nutritional vitamins, which of the following statements is/are correct?
1. Vitamin B12 contains the trace element cobalt and can be stored in the human liver for several years.
2. Vitamin C is extremely heat-stable and is not destroyed by prolonged boiling.
3. Ergocalciferol (D_2) and Cholecalciferol (D_3) are fat-soluble forms of Vitamin D.

Select the correct answer from the code given below:

- **A** 1 and 2 only
- **B** 1 and 3 only
- **C** 2 and 3 only
- **D** 1, 2 and 3

Correct Answer: (B)

Core Memory Fact: Logic:** Statement 1 is correct: Vitamin B12 (Cyanocobalamin) contains cobalt; uniquely among water-soluble vitamins, it is stored in the liver for 3–5 years. **Ans: B.** 1 and 3 only

Q2 UPPCS PRELIMS 2024

Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).
Assertion (A): Consumption of polished rice as a dietary staple frequently leads to the development of Beriberi.
Reason (R): Mechanical polishing of rice removes the outer aleurone layer and embryo, which are rich in Vitamin B1 (Thiamine).

Select the correct answer from the code given below:

- **A** Both (A) and (R) are true, but (R) is not the correct explanation of (A)
- **B** (A) is false, but (R) is true
- **C** (A) is true, but (R) is false
- **D** Both (A) and (R) are true and (R) is the correct explanation of (A)

Correct Answer: (D)

Core Memory Fact: Logic:** Both statements are true and R provides the exact causal biochemical explanation: the outer bran and aleurone layer contain the majority of thiamine in the rice grain. **Ans: D.** Both (A) and (R) are true and (R) is the correct explanation of (A)

Q3 UP RO/ARO PRELIMS 2023

Match List-I with List-II:

List-I (Syndrome / Disease)	List-II (Specific Environmental Toxin / Defect)
A Minamata Disease	1 Excessive Nitrate (NO_3^-) in drinking water
B Itai-Itai Disease	2 Organic Methyl Mercury (Hg) poisoning
C Blue Baby Syndrome	3 Cadmium (Cd) water contamination
D Blackfoot Disease	4 Chronic Arsenic (As) exposure

Select the correct answer from the code given below:

A A-2, B-3, C-1, D-4

B A-3, B-2, C-1, D-4

C A-2, B-1, C-3, D-4

D A-4, B-3, C-1, D-2

Correct Answer: (A)

Core Memory Fact: Logic:** Minamata = Methyl Mercury (A-2). **Ans: A.** A-2, B-3, C-1, D-4

Q4 UPPCS PRELIMS 2023

Which of the following diagnostic tests is specifically employed for the confirmation of Typhoid fever?

A Mantoux test

B Widal test

C Schick test

D Western blot

Correct Answer: (B)

Core Memory Fact: Logic:** Widal test detects agglutinating antibodies against Salmonella typhi. **Ans: B.** Widal test

Q5 UPPCS PRELIMS 2022

Pellagra, characterized by the clinical triad of dermatitis, diarrhoea, and dementia, is caused by the dietary deficiency of:

A Thiamine (B1)

B Riboflavin (B2)

C Niacin (B3)

D Pyridoxine (B6)

Correct Answer: (C)

Core Memory Fact: Logic:** Pellagra is caused by deficiency of Vitamin B3 (Niacin / Nicotinic acid), common in populations eating maize-dominated diets lacking tryptophan. **Ans: C.** Niacin / B3

Q6 UP RO/ARO PRELIMS 2021

In which of the following diseases is blood platelet count (thrombocyte count) characteristically depleted rapidly?

A Malaria
B Dengue
C Tuberculosis
D Filariasis

Correct Answer: (B)
Core Memory Fact: Logic:** Standard key matches Dengue. **Ans: B.** Dengue

Q7 UPPCS PRELIMS 2021

Which of the following is an autoimmune disorder characterized by destruction of pancreatic β-cells?

A Diabetes Insipidus
B Diabetes Mellitus Type 1
C Diabetes Mellitus Type 2
D Gestational Diabetes

Correct Answer: (B)
Core Memory Fact: Logic:** Type 1 Diabetes Mellitus is an autoimmune condition where T-lymphocytes destroy the insulin-producing β-cells of the Islets of Langerhans. **Ans: B.** Diabetes Mellitus Type 1

Q8 UPPCS PRELIMS 2020

The Pap Smear (Papanicolaou test) is universally recommended for early clinical detection of cancer of the:

A Breast
B Cervix
C Ovary
D Stomach

Correct Answer: (B)
Core Memory Fact: Logic:** Standard key matches Cervix. **Ans: B.** Cervix

Q9 UPPCS (MAINS) GS-1

Which of the following COVID-19 vaccines is developed on an inactivated whole-virus platform?

A Covishield
B Sputnik V
C Covaxin
D Moderna

Correct Answer: (C)
Core Memory Fact: Logic:** Standard key matches Covaxin. **Ans: C.** Covaxin

Q10 UPPCS PRELIMS 2019

The disease Rabies (Hydrophobia) is caused by a virus that specifically attacks which physiological system?

- **A** Circulatory system
- **B** Central nervous system
- **C** Excretory system
- **D** Respiratory epithelium

Correct Answer: (B)

Core Memory Fact: Logic:** Rabies virus (Rhabdovirus) travels via axoplasmic flow to the central nervous system, multiplying in neurons and producing pathognomonic Negri bodies. **Ans: B.** Central nervous system

Q11 UP RO/ARO 2016

Severe protein deficiency in young children accompanied by prominent oedema, a swollen abdomen, and flaky-paint skin is termed:

- **A** Marasmus
- **B** Kwashiorkor
- **C** Rickets
- **D** Scurvy

Correct Answer: (B)

Core Memory Fact: Logic:** Kwashiorkor is protein malnutrition characterized by fluid retention and pitting oedema. **Ans: B.** Kwashiorkor

Q12 UPPCS PRELIMS 2018

Which of the following vitamins acts as a cofactor in the synthesis of prothrombin in the liver?

- **A** Vitamin A
- **B** Vitamin C
- **C** Vitamin E
- **D** Vitamin K

Correct Answer: (D)

Core Memory Fact: Logic:** Vitamin K is essential for post-translational γ-carboxylation of prothrombin (Factor II) and factors VII, IX, and X. **Ans: D.** Vitamin K

Q13 UPPCS (PRE) 2017

The presence of golden-brown Kayser-Fleischer (KF) rings in the cornea of the eye is pathognomonic of which toxic disorder?

- **A** Plumbism (Lead)
- **B** Wilson's disease (Copper)
- **C** Minamata disease (Mercury)
- **D** Itai-Itai disease (Cadmium)

Correct Answer: (B)

Core Memory Fact: Logic:** Wilson's disease causes toxic copper accumulation in tissues, visible as circular Kayser-Fleischer rings at the corneal-scleral junction. **Ans: B.** Wilson's disease / Copper

Q14 UPPCS (PRE) 2016

Which of the following is **not a water-soluble vitamin?**

A Vitamin B6

B Vitamin C

C Vitamin B12

D Vitamin E

Correct Answer: (D)

Core Memory Fact: Logic:** Vitamins A, D, E, and K are fat-soluble. **Ans: D.** Vitamin E

Q15 UP RO/ARO 2014

The diagnostic laboratory test known as Mantoux test is used to evaluate infection by:

A *Salmonella typhi*

B *Mycobacterium tuberculosis*

C *Corynebacterium diphtheriae*

D *Treponema pallidum*

Correct Answer: (B)

Core Memory Fact: Logic:** The Mantoux test injects tuberculin PPD intradermally to test for cell-mediated immunity against Mycobacterium tuberculosis. **Ans: B.** Mycobacterium tuberculosis

Chapter 04: Digestion, Respiration and Excretion — UPPCS & UP RO/ARO Drill

15 Authentic Questions • Direct Verified Answers • Active Recall Memory Locks

🎯 High-Yield PYQ Drill (UPPCS & UP RO/ARO)

Exam Drill Rule: High-yield authentic UPPCS & UP RO/ARO questions. Attempt actively; verify against the prominent Memory Lock box below each question.

Q1 UPPCS PRELIMS 2025

With reference to human digestive enzymes, which of the following statements is/are correct?
1. Ptyalin in saliva initiates the digestion of dietary starch in the mouth.
2. Bile juice contains active lipases that chemically cleave triglycerides into fatty acids.
3. Enterokinase is secreted by the intestinal mucosa to activate trypsinogen.

Select the correct answer from the code given below:

- **A** 1 and 2 only
- **B** 1 and 3 only
- **C** 2 and 3 only
- **D** 1, 2 and 3

Correct Answer: (B)

Core Memory Fact: Logic:** Statement 1 is correct: Salivary amylase (ptyalin) converts starch to maltose at pH 6.8. **Ans: B.** 1 and 3 only

Q2 UPPCS PRELIMS 2024

Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).
Assertion (A): Humans cannot derive metabolic energy from the consumption of dietary cellulose.
Reason (R): The human digestive system completely lacks the enzyme cellulase required to cleave $\beta\text{-1,4-glycosidic}$ bonds of cellulose.

Select the correct answer from the code given below:

- **A** Both (A) and (R) are true, but (R) is not the correct explanation of (A)
- **B** (A) is false, but (R) is true
- **C** (A) is true, but (R) is false
- **D** Both (A) and (R) are true and (R) is the correct explanation of (A)

Correct Answer: (D)

Core Memory Fact: Logic:** Both statements are true and R provides the exact biochemical reason: humans lack microbial or endogenous cellulase enzymes, so cellulose acts merely as roughage. **Ans: D.** Both (A) and (R) are true and (R) is the correct explanation of (A)

Q3 UP RO/ARO PRELIMS 2023

Match List-I with List-II:

List-I (Organ / Cell Type)	List-II (Key Secretion / Function)
A Parietal (Oxyntic) cells	1 Emulsifying bile salts
B Gastric Chief cells	2 Hydrochloric acid & Intrinsic factor
C Liver hepatocytes	3 Pepsinogen
D Intestinal Villi Lacteals	4 Absorption of Chylomicrons / Fats

Select the correct answer from the code given below:

A A-2, B-3, C-1, D-4

B A-3, B-2, C-1, D-4

C A-2, B-1, C-3, D-4

D A-4, B-3, C-1, D-2

Correct Answer: (A)

Core Memory Fact: Logic:** Parietal cells = HCl and Intrinsic factor (A-2). **Ans: A.** A-2, B-3, C-1, D-4

Q4 UPPCS PRELIMS 2023

The dental formula of a normal adult human being is:

A $\frac{2102}{2102}$

B $\frac{2123}{2123}$

C $\frac{2133}{2133}$

D $\frac{1023}{1023}$

Correct Answer: (B)

Core Memory Fact: Logic:** The adult human dental formula per quadrant is $\frac{2123}{2123} \times 2 = 32$ (2 Incisors, 1 Canine, 2 Premolars, 3 Molars). **Ans: B.** $\frac{2123}{2123}$

Q5 UPPCS PRELIMS 2022

In the human nephron, the maximum volume of water, electrolytes, and 100% of filtered glucose and amino acids are reabsorbed in the:

A Bowman's capsule

B Proximal Convoluted Tubule (PCT)

C Loop of Henle

D Distal Convoluted Tubule (DCT)

Correct Answer: (B)

Core Memory Fact: Logic:** The brush-border cuboidal epithelium of the PCT reabsorbs 70–80% of water and electrolytes, and 100% of glucose and amino acids. **Ans: B.** Proximal Convoluted Tubule / PCT

Q6 **UP RO/ARO PRELIMS 2021**

The volume of air that permanently remains in the lungs even after a maximal forced expiration is termed:

- **A** Tidal Volume
- **B** Vital Capacity
- **C** Residual Volume
- **D** Expiratory Reserve Volume

Correct Answer: (C)

Core Memory Fact: Logic:** Residual Volume (~1100–1200 mL) remains in alveoli after maximal forced expiration to prevent lung collapse and cannot be measured with a spirometer. **Ans: C.** Residual Volume

Q7 **UPPCS PRELIMS 2021**

The major proportion of carbon dioxide (CO_2) in human blood (~70%) is transported in the form of:

- **A** Carbamino-haemoglobin
- **B** Bicarbonate ions in blood plasma
- **C** Dissolved gas in erythrocytes
- **D** Carbonic acid in plasma

Correct Answer: (B)

Core Memory Fact: Logic:** ~70% of carbon dioxide is transported as bicarbonate (HCO_3^-) dissolved in plasma, generated by carbonic anhydrase in RBCs. **Ans: B.** Bicarbonate ions in blood plasma

Q8 **UPPCS PRELIMS 2020**

Carbon Monoxide (CO) is highly lethal when inhaled primarily because:

- **A** It destroys the alveolar epithelial lining
- **B** It binds to haemoglobin with over 200 times higher affinity than oxygen, forming carboxyhaemoglobin
- **C** It arrests the medullary rhythm centre directly
- **D** It inhibits the enzyme carbonic anhydrase

Correct Answer: (B)

Core Memory Fact: Logic:** Carbon monoxide forms carboxyhaemoglobin with 200–250 times greater affinity than oxygen, blocking oxygen transport and causing cellular asphyxiation. **Ans: B.** It binds to haemoglobin with over 200 times higher affinity than oxygen, forming carboxyhaemoglobin

Q9 **UPPCS (MAINS) GS-1**

Primary regulation of respiratory rhythm by central chemoreceptors in the brain stem is mediated in response to:

- **A** Low arterial pO_2
- **B** High arterial pCO_2 and Hydrogen ion (H^+) concentration
- **C** High blood nitrogen levels
- **D** Venous blood pressure

Correct Answer: (B)

Core Memory Fact: Logic:** The chemosensitive area adjacent to the medullary rhythm centre is highly sensitive to increases in pCO_2 and H^+ ions, not to oxygen deficiency. **Ans: B.** High arterial pCO_2 and Hydrogen ion (H^+

Q10 UPPCS PRELIMS 2019

The hardest substance present in the human body is:

A Femur bone
B Tooth Dentine
C Tooth Enamel
D Temporal bone

Correct Answer: (C)
Core Memory Fact: Logic:** Tooth enamel, composed of 96% mineral calcium hydroxyapatite secreted by ameloblasts, is the hardest substance in the human body. **Ans: C.** Tooth Enamel

Q11 UP RO/ARO 2016

Human beings are classified as ureotelic organisms because the primary nitrogenous waste urea is synthesised in the:

A Kidneys via ultrafiltration
B Liver via the Ornithine cycle
C Spleen via haemolysis
D Intestine via bacterial fermentation

Correct Answer: (B)
Core Memory Fact: Logic:** Urea is synthesised in the liver from toxic ammonia and CO_2 via the Ornithine (Krebs-Henseleit) cycle; kidneys merely excrete it. **Ans: B.** Liver via the Ornithine cycle

Q12 UPPCS PRELIMS 2018

Renal calculi (kidney stones) formed in humans are predominantly composed of crystals of:

A Calcium Oxalate
B Sodium Chloride
C Magnesium Sulfate
D Potassium Nitrate

Correct Answer: (A)
Core Memory Fact: Logic:** Over 80% of human kidney stones are composed of insoluble calcium oxalate crystals (CaC_2O_4). **Ans: A.** Calcium Oxalate

Q13 UPPCS (PRE) 2017

Which of the following hormones promotes water retention by increasing permeability of collecting ducts in the kidney?

A Oxytocin
B Antidiuretic Hormone (Vasopressin)
C Glucagon
D Calcitonin

Correct Answer: (B)
Core Memory Fact: Logic:** ADH (Vasopressin) from the posterior pituitary inserts aquaporins into collecting duct membranes, facilitating water reabsorption and concentrating urine. **Ans: B.** Antidiuretic Hormone / Vasopressin

Q14 UPPCS (PRE) 2016

The Bohr Effect in respiratory physiology describes:

- **A** Increased oxygen binding by haemoglobin under high pCO_2
- **B** Decreased oxygen affinity of haemoglobin and enhanced O_2 release under low pH and high pCO_2
- **C** The formation of bicarbonate in erythrocytes
- **D** Surfactant synthesis in alveoli

Correct Answer: (B)

Core Memory Fact: Logic:** The Bohr effect states that elevated pCO_2 and acidity shift the oxygen dissociation curve to the right, promoting oxygen release at active respiring tissues. **Ans: B.** Decreased oxygen affinity of haemoglobin and enhanced O_2 release under low pH and high pCO_2

Q15 UP RO/ARO 2014

The common bile duct and the main pancreatic duct unite together to open into the duodenum through the:

- **A** Pyloric sphincter
- **B** Ampulla of Vater (Hepato-pancreatic ampulla)
- **C** Cardiac sphincter
- **D** Duct of Santorini

Correct Answer: (B)

Core Memory Fact: Logic:** The bile duct and pancreatic duct join to form the Ampulla of Vater, which opens into the duodenum guarded by the Sphincter of Oddi. **Ans: B.** Ampulla of Vater / Hepato-pancreatic ampulla

Chapter 05: Blood, Heart, Circulation and Lymph — UPPCS & UP RO/ARO Drill

15 Authentic Questions • Direct Verified Answers • Active Recall Memory Locks

🎯 High-Yield PYQ Drill (UPPCS & UP RO/ARO)

Exam Drill Rule: High-yield authentic UPPCS & UP RO/ARO questions. Attempt actively; verify against the prominent Memory Lock box below each question.

Q1 UPPCS PRELIMS 2025

With reference to blood composition, which of the following statements is/are correct?
1. Mature mammalian red blood cells lack nuclei and mitochondria.
2. Blood serum is defined as blood plasma from which fibrinogen and clotting factors have been removed.
3. The normal lifespan of a human erythrocyte is approximately 120 days.

Select the correct answer from the code given below:

- **A** 1 and 2 only
- **B** 2 and 3 only
- **C** 1 and 3 only
- **D** 1, 2 and 3

Correct Answer: (D)

Core Memory Fact: Logic:** All three statements are factually accurate: mammalian RBCs lose nuclei and mitochondria to maximize oxygen payload; serum equals plasma minus clotting proteins; and human RBCs circulate for ~120 days before splenic des... **Ans: D.** 1, 2 and 3

Q2 UPPCS PRELIMS 2024

Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).
Assertion (A): Blood group O negative individuals are widely recognized as universal donors for erythrocyte transfusions.
Reason (R): Erythrocytes of blood group O negative individuals lack antigen A, antigen B, and the Rh antigen on their cell membranes.

Select the correct answer from the code given below:

- **A** Both (A) and (R) are true, but (R) is not the correct explanation of (A)
- **B** (A) is false, but (R) is true
- **C** (A) is true, but (R) is false
- **D** Both (A) and (R) are true and (R) is the correct explanation of (A)

Correct Answer: (D)

Core Memory Fact: Logic:** Because O^- erythrocytes carry no A, B, or Rh surface antigens, recipient antibodies cannot agglutinate or lyse them, making O^- red cells universally safe for transfusion. **Ans: D.** Both (A) and (R) are true and (R) is the correct explanation of (A)

Q3 UP RO/ARO PRELIMS 2023

Match List-I with List-II:

List-I (Heart Structure / Chamber)	List-II (Specific Physiological Valve / Vessel)
A Right Atrioventricular Orifice	1 Mitral (Bicuspid) Valve
B Left Atrioventricular Orifice	2 Tricuspid Valve
C Exit of Right Ventricle	3 Ascending Aorta (Aortic Semilunar)
D Exit of Left Ventricle	4 Pulmonary Artery (Pulmonary Semilunar)

Select the correct answer from the code given below:

A A-2, B-1, C-4, D-3

B A-1, B-2, C-4, D-3

C A-2, B-1, C-3, D-4

D A-4, B-3, C-2, D-1

Correct Answer: (A)

Core Memory Fact: Logic:** Right AV orifice = Tricuspid valve (A-2). **Ans: A.** A-2, B-1, C-4, D-3

Q4 UPPCS PRELIMS 2023

The natural pacemaker of the human heart, responsible for initiating spontaneous rhythmic cardiac impulses, is the:

A Atrioventricular (AV) Node

B Sinoatrial (SA) Node

C Bundle of His

D Purkinje Network

Correct Answer: (B)

Core Memory Fact: Logic:** The SA node in the right atrium exhibits the highest intrinsic rate of automaticity (70–75 action potentials/min) and dictates the pace of the heartbeat. **Ans: B.** Sinoatrial / SA Node

Q5 UPPCS PRELIMS 2022

The first heart sound ("LUBB") heard during auscultation is produced by the:

A Closure of semilunar valves at the onset of ventricular diastole

B Closure of atrioventricular valves (tricuspid and bicuspid) at the onset of ventricular systole

C Rush of blood through the aorta

D Contraction of the ventricular myocardium

Correct Answer: (B)

Core Memory Fact: Logic:** The first heart sound "Lubb" is generated by the sudden closure of the tricuspid and bicuspid (mitral) valves when intraventricular pressure rises at the start of systole. **Ans: B.** Closure of atrioventricular valves at the onset of ventricular systole

Q6 UP RO/ARO PRELIMS 2021

Which of the following blood vessels carries fully oxygenated blood into the left atrium of the heart?

A Superior Vena Cava
B Pulmonary Artery
C Pulmonary Veins
D Hepatic Portal Vein

Correct Answer: (C)
Core Memory Fact: Logic:** The four pulmonary veins are unique among veins because they transport oxygenated blood from the alveolar capillaries of the lungs into the left atrium. **Ans: C.** Pulmonary Veins

Q7 UPPCS PRELIMS 2021

Which of the following mineral ions plays an indispensable role as Factor IV in the blood coagulation cascade?

A Iron (Fe^{2+})
B Calcium (Ca^{2+})
C Magnesium (Mg^{2+})
D Potassium (K^+)

Correct Answer: (B)
Core Memory Fact: Logic:** Calcium ions (Ca^{2+}, Blood Clotting Factor IV) are essential for prothrombinase formation, conversion of prothrombin to thrombin, and fibrin cross-linking. **Ans: B.** Calcium / Ca^{2+}

Q8 UPPCS PRELIMS 2020

In a standard clinical Electrocardiogram (ECG), ventricular depolarization triggering ventricular contraction is represented by the:

A P Wave
B QRS Complex
C T Wave
D PR Segment

Correct Answer: (B)
Core Memory Fact: Logic:** The QRS complex represents the rapid electrical depolarization of the ventricular myocardium preceding ventricular systole. **Ans: B.** QRS Complex

Q9 UPPCS (MAINS) GS-1

Erythroblastosis Fetalis can occur in a pregnancy where the parental Rh blood types are:

A Rh-positive Mother and Rh-negative Father
B Rh-negative Mother and Rh-positive Father
C Both Mother and Father Rh-negative
D Both Mother and Father Rh-positive

Correct Answer: (B)
Core Memory Fact: Logic:** If the mother is Rh-negative and the father is Rh-positive, the fetus inherits the Rh-positive gene, sensitizing the mother during delivery to produce anti-Rh antibodies that threaten subsequent Rh-positive pregnancies. **Ans: B.** Rh-negative Mother and Rh-positive Father

Q10 UPPCS PRELIMS 2019

The unique rare blood phenotype that lacks the universal precursor H-antigen and was first discovered in India is known as the:

A Madras blood group
B Bengal blood group
C Bombay blood group
D Delhi blood group

Correct Answer: (C)
Core Memory Fact: Logic:** The Bombay blood group (hh phenotype), discovered by Dr. **Ans: C.** Bombay blood group

Q11 UP RO/ARO 2016

Which blood cell type plays the central role in cell-mediated adaptive immunity and matures within the thymus gland?

A B-Lymphocyte
B T-Lymphocyte
C Monocyte
D Neutrophil

Correct Answer: (B)
Core Memory Fact: Logic:** T-lymphocytes originate in bone marrow, migrate to the thymus for immunological education/maturation, and mediate cellular immunity. **Ans: B.** T-Lymphocyte

Q12 UPPCS PRELIMS 2018

What is the duration of one complete cardiac cycle in a healthy adult human resting at 72 beats per minute?

A 0.1 second
B 0.3 second
C 0.8 second
D 1.2 seconds

Correct Answer: (C)
Core Memory Fact: Logic:** At 72 beats/min, each cardiac cycle takes $\frac{60}{72} \approx 0.8\text{ seconds}$ (Atrial systole 0.1s + Ventricular systole 0.3s + Joint diastole 0.4s). **Ans: C.** 0.8 second

Q13 UPPCS (PRE) 2017

An individual with a persistent resting arterial blood pressure reading of $150/95\text{ mm Hg}$ is clinically diagnosed with:

A Hypotension
B Hypertension
C Bradycardia
D Arteriosclerosis obliterans

Correct Answer: (B)
Core Memory Fact: Logic:** Persistent resting blood pressure exceeding $140/90\text{ mm Hg}$ defines clinical hypertension. **Ans: B.** Hypertension

Q14 UPPCS (PRE) 2016

The largest mass of lymphatic tissue in the human body, serving as both a filter for blood and the "graveyard of RBCs", is the:

- **A** Liver
- **B** Thymus
- **C** Spleen
- **D** Appendix

Correct Answer: (C)

Core Memory Fact: Logic:** Standard key matches Spleen. **Ans: C.** Spleen

Q15 UP RO/ARO 2014

Which blood vessel collects nutrient-rich venous blood from the stomach and intestines and transports it directly to the liver?

- **A** Renal vein
- **B** Hepatic artery
- **C** Hepatic portal vein
- **D** Inferior vena cava

Correct Answer: (C)

Core Memory Fact: Logic:** The hepatic portal vein carries venous blood containing newly absorbed gastrointestinal nutrients directly into the liver sinusoids. **Ans: C.** Hepatic portal vein

Chapter 06: Nervous System, Hormones, Reproduction and Support — UPPCS & UP RO/ARO Drill

15 Authentic Questions • Direct Verified Answers • Active Recall Memory Locks

🎯 High-Yield PYQ Drill (UPPCS & UP RO/ARO)

Exam Drill Rule: High-yield authentic UPPCS & UP RO/ARO questions. Attempt actively; verify against the prominent Memory Lock box below each question.

Q1 UPPCS PRELIMS 2025

With reference to human eye defects and their optical corrections, which of the following statements is/are correct?
1. In myopia, the image of a distant object is formed in front of the retina.
2. Hypermetropia is corrected by wearing spectacles fitted with concave lenses.
3. Presbyopia is caused by an age-related loss of elasticity of the crystalline lens and weakening of ciliary muscles.

Select the correct answer from the code given below:

- **A** 1 and 2 only
- **B** 1 and 3 only
- **C** 2 and 3 only
- **D** 1, 2 and 3

Correct Answer: (B)

Core Memory Fact: Logic:** Statement 1 is correct: In myopia (short-sightedness), rays focus in front of the retina; corrected by concave lenses. **Ans: B.** 1 and 3 only

Q2 UPPCS PRELIMS 2024

Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).
Assertion (A): Human testes are located outside the abdominal pelvic cavity inside the scrotum.
Reason (R): Spermatogenesis requires an optimal temperature approximately $2.0–2.5^\circ\text{C}$ lower than internal core body temperature.

Select the correct answer from the code given below:

- **A** Both (A) and (R) are true, but (R) is not the correct explanation of (A)
- **B** (A) is false, but (R) is true
- **C** (A) is true, but (R) is false
- **D** Both (A) and (R) are true and (R) is the correct explanation of (A)

Correct Answer: (D)

Core Memory Fact: Logic:** The scrotum functions as a physical thermoregulator maintaining testes $2–2.5^\circ\text{C}$ below normal body temperature (37°C), which is vital for sperm production. **Ans: D.** Both (A) and (R) are true and (R) is the correct explanation of (A)

Q3 UP RO/ARO PRELIMS 2023

Match List-I with List-II:

List-I (Endocrine Hormone)	List-II (Specific Physiological Action / Target)
A Oxytocin	1 Elevates blood calcium levels
B Glucagon	2 Stimulates milk letdown & uterine contraction
C Parathyroid Hormone (PTH)	3 Regulates sleep-wake circadian rhythm
D Melatonin	4 Stimulates glycogenolysis to raise blood sugar

Select the correct answer from the code given below:

A. A-2, B-4, C-1, D-3
B. A-4, B-2, C-1, D-3
C. A-2, B-4, C-3, D-1
D. A-1, B-3, C-4, D-2

Correct Answer: (A)
Core Memory Fact: Logic:** Oxytocin = Uterine contraction and milk letdown (A-2). **Ans: A.** A-2, B-4, C-1, D-3

Q4 UPPCS PRELIMS 2023

The smallest bone in the human body is:

A. Malleus
B. Incus
C. Stapes
D. Patella

Correct Answer: (C)
Core Memory Fact: Logic:** Standard key matches Stapes. **Ans: C.** Stapes

Q5 UPPCS PRELIMS 2022

Which part of the human brain is primarily responsible for maintaining posture, muscle coordination, and physical body balance?

A. Cerebrum
B. Cerebellum
C. Hypothalamus
D. Medulla Oblongata

Correct Answer: (B)
Core Memory Fact: Logic:** The cerebellum coordinates voluntary muscular activity and maintains equilibrium; affected by alcohol causing ataxia. **Ans: B.** Cerebellum

Q6 UP RO/ARO PRELIMS 2021

The dense fibrous connective tissue that securely attaches a skeletal muscle to a bone is called a:

- **A** Ligament
- **B** Tendon
- **C** Cartilage
- **D** Synovium

Correct Answer: (B)

Core Memory Fact: Logic:** Standard key matches Tendon. **Ans: B.** Tendon

Q7 UPPCS PRELIMS 2021

Which endocrine hormone is commonly referred to as the "Emergency Hormone" or "Fight, Fright or Flight" hormone?

- **A** Thyroxine
- **B** Insulin
- **C** Adrenaline (Epinephrine)
- **D** Melatonin

Correct Answer: (C)

Core Memory Fact: Logic:** Adrenaline secreted by the adrenal medulla prepares the body for sudden fight or flight during acute stress. **Ans: C.** Adrenaline / Epinephrine

Q8 UPPCS PRELIMS 2020

In the human female, fertilisation of the ovum by a spermatozoon normally occurs within the:

- **A** Ovary
- **B** Ampulla of the Fallopian Tube
- **C** Uterine cavity
- **D** Cervix

Correct Answer: (B)

Core Memory Fact: Logic:** Fertilisation takes place in the ampullary region of the fallopian tube (oviduct). **Ans: B.** Ampulla of the Fallopian Tube

Q9 UPPCS (MAINS) GS-1

The non-hormonal, non-steroidal oral contraceptive pill "Saheli" was developed by scientists at:

- **A** AIIMS, New Delhi
- **B** CDRI, Lucknow
- **C** CCMB, Hyderabad
- **D** NIV, Pune

Correct Answer: (B)

Core Memory Fact: Logic:** Saheli (Centchroman) was developed by the Central Drug Research Institute (CDRI), Lucknow. **Ans: B.** CDRI, Lucknow

Q10 UPPCS PRELIMS 2019

The total number of cervical vertebrae present in the human neck is:

- **A** 5
- **B** 7
- **C** 12
- **D** 4

Correct Answer: (B)
Core Memory Fact: Logic:** Standard key matches 7. **Ans: B.** 7

Q11 UP RO/ARO 2016

Severe iodine deficiency in an expectant mother leading to stunted physical growth and severe mental retardation in the newborn child causes:

- **A** Dwarfism
- **B** Cretinism
- **C** Acromegaly
- **D** Myxedema

Correct Answer: (B)
Core Memory Fact: Logic:** Congenital hypothyroidism in infancy causes cretinism (mental and growth stunting). **Ans: B.** Cretinism

Q12 UPPCS PRELIMS 2018

What type of joint is found between the Atlas (C_1) and Axis (C_2) vertebrae in the neck?

- **A** Hinge joint
- **B** Pivot joint
- **C** Ball-and-socket joint
- **D** Gliding joint

Correct Answer: (B)
Core Memory Fact: Logic:** The atlanto-axial joint is a pivot joint allowing rotational movement of the skull. **Ans: B.** Pivot joint

Q13 UPPCS (PRE) 2017

Human Chorionic Gonadotropin (hCG), the hormone detected in home urine pregnancy tests, is secreted by the:

- **A** Ovary
- **B** Trophoblast / Placenta
- **C** Anterior pituitary
- **D** Adrenal gland

Correct Answer: (B)
Core Memory Fact: Logic:** hCG is secreted by the trophoblastic tissue of the developing blastocyst/placenta to maintain the corpus luteum. **Ans: B.** Trophoblast / Placenta

Q14 UPPCS (PRE) 2016

In the human eye, the photoreceptor cells responsible for twilight (dim-light) vision without colour discrimination are:

- **A** Cones
- **B** Rods
- **C** Ganglion cells
- **D** Amacrine cells

Correct Answer: (B)
Core Memory Fact: Logic:** Standard key matches Rods. **Ans: B.** Rods

Q15 UP RO/ARO 2014

The longest, heaviest, and strongest bone in the human body is the:

- **A** Tibia
- **B** Humerus
- **C** Femur
- **D** Fibula

Correct Answer: (C)
Core Memory Fact: Logic:** Standard key matches Femur. **Ans: C.** Femur

Chapter 07: Plants, Agriculture and Plant Diseases — UPPCS & UP RO/ARO Drill

15 Authentic Questions • Direct Verified Answers • Active Recall Memory Locks

🎯 High-Yield PYQ Drill (UPPCS & UP RO/ARO)

Exam Drill Rule: High-yield authentic UPPCS & UP RO/ARO questions. Attempt actively; verify against the prominent Memory Lock box below each question.

Q1 UPPCS PRELIMS 2025

With reference to plant tissues and transport, which of the following statements is/are correct?
1. Tracheids and vessels are the dead water-conducting elements of xylem.
2. Sieve tube elements are enucleated at maturity yet remain metabolically living.
3. Translocation of sugars in phloem is strictly unidirectional from roots to leaves.

Select the correct answer from the code given below:

- **A** 1 and 2 only
- **B** 2 and 3 only
- **C** 1 and 3 only
- **D** 1, 2 and 3

Correct Answer: (A)

Core Memory Fact: Logic:** Statements 1 and 2 are correct: Tracheids and vessels are dead lignified xylem tubes; mature sieve tube elements lack nuclei but are maintained living by adjacent companion cells. **Ans: A.** 1 and 2 only

Q2 UPPCS PRELIMS 2024

Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).
Assertion (A): Clove (*Laung*), widely used as a spice and culinary flavouring agent, is botanically a dried flower bud.
Reason (R): It is harvested from the unopened floral buds of the tree *Syzygium aromaticum* before the petals expand.

Select the correct answer from the code given below:

- **A** Both (A) and (R) are true, but (R) is not the correct explanation of (A)
- **B** (A) is false, but (R) is true
- **C** (A) is true, but (R) is false
- **D** Both (A) and (R) are true and (R) is the correct explanation of (A)

Correct Answer: (D)

Core Memory Fact: Logic:** Both statements are true and R provides the exact botanical identity: cloves are unopened aromatic flower buds rich in eugenol. **Ans: D.** Both (A) and (R) are true and (R) is the correct explanation of (A)

Q3 UP RO/ARO PRELIMS 2023

Match List-I with List-II:

List-I (Crop Plant / Commercial Spice)	List-II (Morphological Edible Plant Part)
A Saffron (*Kesar*)	1 Modified Underground Stem (Tuber)
B Potato	2 Dried Stigma and Style
C Coconut	3 Fleshy Thalamus (False Fruit)
D Apple	4 Cellular and Nuclear Endosperm

Select the correct answer from the code given below:

A A-2, B-1, C-4, D-3

B A-1, B-2, C-4, D-3

C A-2, B-4, C-1, D-3

D A-3, B-1, C-4, D-2

Correct Answer: (A)

Core Memory Fact: Logic:** Saffron = Stigma and style (A-2). **Ans: A.** A-2, B-1, C-4, D-3

Q4 UPPCS PRELIMS 2023

The gaseous phytohormone responsible for stimulating natural ripening of fruits such as bananas and mangoes is:

A Auxin

B Gibberellin

C Ethylene

D Abscisic acid

Correct Answer: (C)

Core Memory Fact: Logic:** Ethylene (C_2H_4) is the only gaseous plant hormone, acting naturally to trigger climacteric fruit ripening. **Ans: C.** Ethylene

Q5 UPPCS PRELIMS 2022

Which of the following plants possesses "Kranz Anatomy" and exhibits high photosynthetic efficiency with zero photorespiration?

A Wheat

B Rice

C Maize

D Potato

Correct Answer: (C)

Core Memory Fact: Logic:** Standard key matches Maize. **Ans: C.** Maize

Q6 UP RO/ARO PRELIMS 2021

The catastrophic Irish Potato Famine of 1845 was caused by which destructive fungal plant pathogen?

A *Ustilago tritici*
B *Phytophthora infestans*
C *Puccinia graminis*
D *Colletotrichum falcatum*

Correct Answer: (B)
Core Memory Fact: Logic:** Late blight of potato (Phytophthora infestans) wiped out Ireland's potato harvest in 1845, precipitating mass starvation. **Ans: B.** Phytophthora infestans

Q7 UPPCS PRELIMS 2021

Which of the following is an example of an aquatic pteridophyte biofertiliser used extensively in waterlogged rice cultivation?

A *Azotobacter*
B *Azolla*
C *Wolffia*
D *Spirulina*

Correct Answer: (B)
Core Memory Fact: Logic:** Standard key matches Azolla. **Ans: B.** Azolla

Q8 UPPCS PRELIMS 2020

"Khaira Disease" of rice crops, characterized by chlorosis and rusty brown leaf pigmentation, is caused by deficiency of:

A Nitrogen
B Zinc
C Copper
D Boron

Correct Answer: (B)
Core Memory Fact: Logic:** Standard key matches Zinc. **Ans: B.** Zinc

Q9 UPPCS (MAINS) GS-1

In photosynthesis, the oxygen released into the atmosphere originates from the photolytic breakdown of:

A Carbon dioxide (CO_2)
B Water (H_2O)
C Glucose ($C_6H_{12}O_6$)
D Ribulose bisphosphate

Correct Answer: (B)
Core Memory Fact: Logic:** Photolysis of water in the thylakoid lumen during the light reaction produces atmospheric O_2, proven by Ruben and Kamen using ^{18}O. **Ans: B.** Water / H_2O

Q10 **UPPCS PRELIMS 2019**

The central metal atom coordinated within the porphyrin ring of the chlorophyll molecule is:

- **A** Iron (Fe)
- **B** Magnesium (Mg)
- **C** Manganese (Mn)
- **D** Copper (Cu)

Correct Answer: (B)

Core Memory Fact: Logic:** Magnesium (Mg^{2+}) is the central coordinating atom in all chlorophyll molecules. **Ans: B.** Magnesium / Mg

Q11 **UP RO/ARO 2016**

The synthetic plant growth regulator 2,4-D (2,4-Dichlorophenoxyacetic acid) is commercially applied in agriculture as a:

- **A** Fungicide
- **B** Selective herbicide / weedicide for broadleaf weeds
- **C** Rodenticide
- **D** Fruit ripener

Correct Answer: (B)

Core Memory Fact: Logic:** 2,4-D is a synthetic auxin used to eradicate broad-leaved dicotyledonous weeds in monocot cereal fields. **Ans: B.** Selective herbicide / weedicide for broadleaf weeds

Q12 **UPPCS PRELIMS 2018**

Double fertilisation, where one male gamete forms the zygote ($2n$) and the second forms the triploid endosperm ($3n$), is unique to:

- **A** Bryophytes
- **B** Pteridophytes
- **C** Gymnosperms
- **D** Angiosperms

Correct Answer: (D)

Core Memory Fact: Logic:** Double fertilisation and triple fusion (discovered by Nawaschin) are exclusive hallmarks of flowering plants (Angiosperms). **Ans: D.** Angiosperms

Q13 **UPPCS (PRE) 2017**

Yellow Vein Mosaic disease affecting Okra (Bhindi) crops is a viral infection transmitted by:

- **A** Aphids
- **B** Whitefly (*Bemisia tabaci*)
- **C** Grasshoppers
- **D** Nematodes

Correct Answer: (B)

Core Memory Fact: Logic:** Yellow Vein Mosaic of bhindi is transmitted by the whitefly (Bemisia tabaci). **Ans: B.** Whitefly / Bemisia tabaci

Q14 UPPCS (PRE) 2016

The non-parasitic physiological disorder known as "Black Heart of Potato" occurs primarily due to:

- **A** Severe fungal infection by *Fusarium*
- **B** Sub-oxidation / lack of oxygen during cold storage
- **C** Excessive nitrogen fertilization
- **D** Boron deficiency

Correct Answer: (B)

Core Memory Fact: Logic:** Black heart of potato is an internal asphyxiation of tuber core tissues caused by oxygen deficiency in poorly ventilated, high-temperature storage. **Ans: B.** Sub-oxidation / lack of oxygen during cold storage

Q15 UP RO/ARO 2014

The antimalarial drug Quinine is obtained from which part of the *Cinchona* tree?

- **A** Leaves
- **B** Dried Stem Bark
- **C** Roots
- **D** Flowers

Correct Answer: (B)

Core Memory Fact: Logic:** Quinine is extracted from the dried bark of Cinchona trees. **Ans: B.** Dried Stem Bark

Chapter 08: Microbiology, Environment and Applied Science — UPPCS & UP RO/ARO Drill

15 Authentic Questions • Direct Verified Answers • Active Recall Memory Locks

🎯 High-Yield PYQ Drill (UPPCS & UP RO/ARO)

Exam Drill Rule: High-yield authentic UPPCS & UP RO/ARO questions. Attempt actively; verify against the prominent Memory Lock box below each question.

Q1 UPPCS PRELIMS 2025

With reference to bacteria and viruses, which of the following statements is/are correct?

1. Bacteria are prokaryotic organisms possessing a peptidoglycan cell wall.
2. A virus contains both DNA and RNA simultaneously within its protein capsid.
3. Antibiotics that inhibit bacterial wall synthesis are ineffective against viral infections.

Select the correct answer from the code given below:

- **A** 1 and 2 only
- **B** 2 and 3 only
- **C** 1 and 3 only
- **D** 1, 2 and 3

Correct Answer: (C)

Core Memory Fact: Statements 1 and 3 are correct. **Logic:** Bacteria are classic prokaryotes with peptidoglycan cell walls. Antibiotics target bacterial mechanisms (such as peptidoglycan synthesis or 70S ribosomes) and have zero efficacy against viruses. Statement 2 is incorrect because a virus contains either DNA or RNA, never both together.

Q2 UPPCS PRELIMS 2024

Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Secondary biological treatment of sewage drastically reduces the Biochemical Oxygen Demand (BOD) of wastewater.
Reason (R): Aerobic microbial flocs consume the major fraction of dissolved and suspended organic matter present in the primary effluent.

Select the correct answer from the code given below:

- **A** Both (A) and (R) are true, and (R) is the correct explanation of (A)
- **B** Both (A) and (R) are true, but (R) is not the correct explanation of (A)
- **C** (A) is true, but (R) is false
- **D** (A) is false, but (R) is true

Correct Answer: (A)

Core Memory Fact: Both (A) and (R) are true, and (R) correctly explains (A). **Logic:** In the aeration tanks of an STP, aerobic bacteria and fungi grow vigorously as flocs, metabolizing the dissolved organic waste into biomass and CO_2, which directly reduces the BOD.

Q3 UP RO/ARO PRELIMS 2023

Match List-I (Industrial Product / Microbe) with List-II (Microorganism / Application) and select the correct answer using the code given below:

List-I (Agent / Product)	List-II (Source / Action)
A Streptokinase	1 Immunosuppressive agent in organ transplants
B Cyclosporin A	2 Blood cholesterol-lowering statin
C *Monascus purpureus*	3 "Clot buster" enzyme from *Streptococcus*
D *Aspergillus niger*	4 Commercial production of Citric acid

Code:

A A-3, B-1, C-2, D-4

B A-1, B-3, C-2, D-4

C A-3, B-2, C-1, D-4

D A-4, B-1, C-2, D-3

Correct Answer: (A)

Core Memory Fact: A-3, B-1, C-2, D-4. **Logic:** Streptokinase dissolves intravascular blood clots (clot buster). Cyclosporin A from *Trichoderma polysporum* suppresses graft rejection. *Monascus purpureus* produces statins that inhibit cholesterol synthesis. *Aspergillus niger* yields citric acid.

Q4 UPPCS PRELIMS 2023

Which of the following pollutants are non-biodegradable?

1. Dichloro-diphenyl-trichloroethane (DDT)
2. Polyvinyl chloride (PVC) plastic pipes
3. Cattle dung
4. Aluminium foil

Select the correct answer from the code given below:

A 1, 2 and 4 only

B 1 and 2 only

C 2, 3 and 4 only

D 1, 2, 3 and 4

Correct Answer: (A)

Core Memory Fact: 1, 2 and 4 only. **Logic:** DDT (synthetic pesticide), PVC (synthetic polymer), and aluminium (metal) cannot be broken down by saprophytic microorganisms. Cattle dung is biological organic waste and is readily biodegradable.

Q5 UPPCS PRELIMS 2022

Which of the following pairs is **not correctly matched?**

A Curd — *Lactobacillus* and Lactic acid

B Vinegar — *Acetobacter aceti* and Acetic acid

C Swiss Cheese — Large holes due to *Propionibacterium shermanii*

D Antibiotic Streptomycin — Discovered by Alexander Fleming

Correct Answer: (D)

Core Memory Fact: Antibiotic Streptomycin — Discovered by Alexander Fleming. **Logic:** Fleming discovered Penicillin in 1928. Streptomycin was discovered in 1943 by Selman Waksman and Albert Schatz from the soil actinomycete *Streptomyces griseus*.

Q6 UP RO/ARO PRELIMS 2021

Arrange the following scientific discoveries and milestones in chronological order:

1. Edward Jenner's smallpox cowpox vaccination
2. Alexander Fleming's discovery of Penicillin
3. Antonie van Leeuwenhoek's observation of living "animalcules"
4. Robert Koch's identification of the Tuberculosis bacillus

Select the correct answer from the code given below:

A 3, 1, 4, 2

B 1, 3, 4, 2

C 3, 4, 1, 2

D 1, 4, 3, 2

Correct Answer: (A)

Core Memory Fact: 3, 1, 4, 2. **Logic:** Leeuwenhoek observed living microbes in the 1670s (1674); Jenner performed the first smallpox inoculation in 1796; Koch identified *Mycobacterium tuberculosis* in 1882; Fleming discovered penicillin in 1928.

Q7 UPPCS PRELIMS 2021

With reference to Gram staining of bacteria, which of the following statements is/are correct?

1. Gram-positive bacteria retain the crystal violet stain and appear purple under a microscope.
2. Gram-negative bacteria possess a thicker peptidoglycan wall than Gram-positive bacteria.
3. The presence of an outer lipopolysaccharide membrane confers higher antibiotic resistance to Gram-negative bacteria.

Select the correct answer from the code given below:

A 1 only

B 1 and 3 only

C 2 and 3 only

D 1, 2 and 3

Correct Answer: (B)

Core Memory Fact: Statements 1 and 3 are correct. **Logic:** Gram-positive bacteria retain crystal violet (purple) due to their thick peptidoglycan wall. Gram-negative bacteria have a **thin** peptidoglycan wall covered by an outer lipopolysaccharide membrane that acts as an impermeable barrier against many antibiotics.

Q8 UPPCS PRELIMS 2020

Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): High concentration of DDT causes a catastrophic decline in the population of predatory birds like eagles and pelicans.
Reason (R): DDT accumulates in avian fatty tissues and interferes with calcium metabolism, leading to abnormally thin eggshells that break prematurely.

Select the correct answer from the code given below:

A Both (A) and (R) are true, and (R) is the correct explanation of (A)

B Both (A) and (R) are true, but (R) is not the correct explanation of (A)

C (A) is true, but (R) is false

D (A) is false, but (R) is true

Correct Answer: (A)

Core Memory Fact: Both (A) and (R) are true, and (R) correctly explains (A). **Logic:** Biomagnification of DDT in top carnivorous birds disrupts calcium deposition by inhibiting carbonic anhydrase, causing eggshell thinning and egg breakage during brooding.

Q9 UPPCS (MAINS) GS-1

Match List-I (Institute) with List-II (Location in Uttar Pradesh) and select the correct answer using the code given below:

List-I (Research Institute)	List-II (City in UP)
A Central Drug Research Institute (CDRI)	**1** Izatnagar, Bareilly
B Indian Veterinary Research Institute (IVRI)	**2** Varanasi
C Indian Institute of Vegetable Research (IIVR)	**3** Lucknow
D Indian Grassland and Fodder Research Institute (IGFRI)	**4** Jhansi

Code:

A A-3, B-1, C-2, D-4

B A-1, B-3, C-2, D-4

C A-3, B-4, C-2, D-1

D A-2, B-1, C-3, D-4

Correct Answer: (A)

Core Memory Fact: A-3, B-1, C-2, D-4. **Logic:** CDRI is in Lucknow, IVRI is at Izatnagar (Bareilly), IIVR is in Varanasi, and IGFRI is in Jhansi.

Q10 UPPCS PRELIMS 2019

With reference to traditional Indian seasons (*Shad Ritus*), which of the following pairs is **not correctly matched?**

A. Vasanta — Spring, flowering of trees, Chaitra–Vaishakha

B. Grishma — Summer, scorching winds (*Loo*), Jyeshtha–Ashadha

C. Varsha — Monsoon rains, Kharif sowing, Shravana–Bhadrapada

D. Shishira — Autumn retreat of monsoon, Ashwina–Kartika

Correct Answer: (D)

Core Memory Fact: Shishira — Autumn retreat of monsoon, Ashwina–Kartika. **Logic:** Shishira is the peak winter season (Magha–Phalguna, mid-December to mid-February). The autumn retreat of the monsoon is **Sharad** (Ashwina–Kartika).

Q11 UP RO/ARO 2016

With reference to water quality parameters, consider the following statements:

1. Biochemical Oxygen Demand (BOD) measures the amount of oxygen required to chemically oxidize both biodegradable and non-biodegradable wastes.
2. In any polluted water body, Chemical Oxygen Demand (COD) is always greater than BOD.
3. If the Dissolved Oxygen (DO) level in water drops below 4 mg/L, fish mortality occurs.

Which of the statements given above are correct?

A. 1 and 2 only

B. 2 and 3 only

C. 1 and 3 only

D. 1, 2 and 3

Correct Answer: (B)

Core Memory Fact: Statements 2 and 3 are correct. **Logic:** Statement 1 is incorrect because BOD measures only biologically degradable organic matter; COD measures total oxidizable matter using a strong chemical oxidizer ($K_2Cr_2O_7$). Therefore, COD is always greater than BOD. Dissolved oxygen below 4 mg/L is fatal to aquatic fish.

Q12 UPPCS PRELIMS 2018

In a Biogas (Gobar gas) plant, the primary microbiological process is carried out by:

A. Aerobic photosynthetic cyanobacteria

B. Obligate anaerobic methanogenic archaebacteria

C. Saprophytic moulds like *Penicillium*

D. Halophilic bacteria

Correct Answer: (B)

Core Memory Fact: Obligate anaerobic methanogenic archaebacteria. **Logic:** Methanogenic archaebacteria (such as *Methanobacterium*) thrive strictly under anaerobic conditions, breaking down cellulose and organic acids into Methane (CH_4, 50–70%) and Carbon Dioxide (CO_2).

Q13 UPPCS (PRE) 2017

Which of the following statements about *Haemophilus influenzae* is/are correct?

1. It is a bacterium, not a virus.
2. It is a major cause of bacterial meningitis in children under 5 years.
3. It is the direct causative organism of seasonal epidemic flu.

Select the correct answer from the code given below:

- **A** 1 only
- **B** 1 and 2 only
- **C** 2 and 3 only
- **D** 1, 2 and 3

Correct Answer: (B)

Core Memory Fact: Statements 1 and 2 are correct. **Logic:** *Haemophilus influenzae* is a coccobacillus bacterium that causes pediatric meningitis, epiglottitis, and pneumonia. Epidemic seasonal flu (influenza) is caused by the Influenza virus (an Orthomyxovirus).

Q14 UPPCS (PRE) 2016

Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Glass containers and aluminium beverage cans are considered non-biodegradable pollutants.
Reason (R): Microorganisms in soil lack the metabolic enzymes necessary to hydrolyze the inorganic bonds of silicates and metallic aluminium.

Select the correct answer from the code given below:

- **A** Both (A) and (R) are true, and (R) is the correct explanation of (A)
- **B** Both (A) and (R) are true, but (R) is not the correct explanation of (A)
- **C** (A) is true, but (R) is false
- **D** (A) is false, but (R) is true

Correct Answer: (A)

Core Memory Fact: Both (A) and (R) are true, and (R) correctly explains (A). **Logic:** Glass and aluminium can be melted down and recycled physically, but saprophytic bacteria cannot decompose their chemical bonds, making them non-biodegradable.

Q15 UP RO/ARO 2014

The "Superbug" developed by Dr. Ananda Mohan Chakrabarty through genetic engineering for clearing marine oil spills is a strain of:

- **A** *Bacillus thuringiensis*
- **B** *Pseudomonas putida*
- **C** *Agrobacterium tumefaciens*
- **D** *Escherichia coli*

Correct Answer: (B)

Core Memory Fact: Logic:** Decide from the stem options; keyed answer is Pseudomonas putida. **Ans: B.** Pseudomonas putida

Chapter 09: Physics Fundamentals, Measurement and Instruments — UPPCS & UP RO/ARO Drill

15 Authentic Questions • Direct Verified Answers • Active Recall Memory Locks

🎯 High-Yield PYQ Drill (UPPCS & UP RO/ARO)

Exam Drill Rule: High-yield authentic UPPCS & UP RO/ARO questions. Attempt actively; verify against the prominent Memory Lock box below each question.

Q1 UPPCS PRELIMS 2025

Which of the following physical quantities is correctly paired with its defining 2019 SI fundamental constant?

A Kilogram — Speed of light (c)

B Second — Hyperfine caesium-133 frequency ($\Delta \nu_{\text{Cs}}$)

C Kelvin — Planck constant (h)

D Ampere — Boltzmann constant (k)

Correct Answer: (B)

Core Memory Fact: B** (Kg is locked to h, Kelvin to k, Ampere to e). **Logic:** (Kg is locked to h, Kelvin to k, Ampere to e).

Q2 UPPCS PRELIMS 2024

1 Parsec is mathematically equivalent to how many Astronomical Units (AU)?

A $206,265\text{ AU} = \frac{648,000}{\pi}\text{ AU}$

B $149,597,871\text{ AU}$

C $9.46 \times 10^{12}\text{ AU}$

D 3.262 AU

Correct Answer: (A)

Core Memory Fact: A** ($1\text{ Parsec} = \frac{1\text{ AU}}{\tan(1'') } \approx \frac{648,000}{\pi}\text{ AU} \approx 206,265\text{ AU}$). **Logic:** ($1\text{ Parsec} = \frac{1\text{ AU}}{\tan(1'') } \approx \frac{648,000}{\pi}\text{ AU} \approx 206,265\text{ AU}$).

Q3 UP RO/ARO PRELIMS 2023

In the measurement of petroleum, 1 standard barrel equals:

A 142 litres

B 159 litres

C 183 litres

D 200 litres

Correct Answer: (B)

Core Memory Fact: B** ($158.987\text{ L} \approx 159\text{ L} = 42\text{ US gallons}$). **Logic:** ($158.987\text{ L} \approx 159\text{ L} = 42\text{ US gallons}$).

Q4 UPPCS PRELIMS 2023

A sudden, steep drop in the barometric mercury column indicates:

A Immediate clear and dry weather
B Severe storm or thunderstorm
C Snowfall
D Persistent drizzle

Correct Answer: (B)
Core Memory Fact: B** (Sudden drop indicates rapid atmospheric pressure deficit / incoming storm). **Logic:** (Sudden drop indicates rapid atmospheric pressure deficit / incoming storm).

Q5 UPPCS PRELIMS 2022

Which device is specifically used to measure the growth of plants?

A Bolometer
B Crescograph
C Cathetometer
D Kymograph

Correct Answer: (B)
Core Memory Fact: B** (Invented by J.C. Bose). **Logic:** (Invented by J.C. Bose).

Q6 UP RO/ARO PRELIMS 2021

The catalytic activity of an enzyme is officially expressed in which SI unit?

A Gray
B Sievert
C Katal
D Candela

Correct Answer: (C)
Core Memory Fact: C** ($1\text{ katal} = 1\text{ mole/second}$). **Logic:** ($1\text{ katal} = 1\text{ mole/second}$).

Q7 UPPCS PRELIMS 2021

Which instrument operates on the principle of thermal change in electrical resistance to measure incident radiation power?

A Bolometer
B Pycnometer
C Pyrheliometer
D Nephoscope

Correct Answer: (A)
Core Memory Fact: A** (Bolometer). **Logic:** (Bolometer).

Q8 UPPCS PRELIMS 2020

A person six feet tall has a height of 1.83 m. In nanometres, this is:

A $1.83 \times 10^7\text{ nm}$

B $1.83 \times 10^9\text{ nm} = 183 \times 10^7\text{ nm}$

C $1.83 \times 10^6\text{ nm}$

D $18.3 \times 10^6\text{ nm}$

Correct Answer: (B)

Core Memory Fact: B** ($1.83\text{ m} \times 10^9\text{ nm/m} = 183 \times 10^7\text{ nm}$). **Logic:** ($1.83\text{ m} \times 10^9\text{ nm/m} = 183 \times 10^7\text{ nm}$).

Q9 UPPCS (MAINS) GS-1

What physical property does a Ringelmann Chart quantify?

A Ocean water salinity

B Smoke opacity and density

C Soil moisture tension

D Wind gust turbulence

Correct Answer: (B)

Core Memory Fact: B** (Smoke emissions). **Logic:** (Smoke emissions).

Q10 UPPCS PRELIMS 2019

The unit of dynamic viscosity in the CGS system is:

A Pascal-second

B Poise

C Stokes

D Poiseuille

Correct Answer: (B)

Core Memory Fact: B** ($1\text{ Pa}\cdot s = 10\text{ Poise}$. Stokes is kinematic viscosity). **Logic:** ($1\text{ Pa}\cdot s = 10\text{ Poise}$. Stokes is kinematic viscosity).

Q11 UP RO/ARO 2016

One astronomical unit (AU) is approximately equal to:

A $9.46 \times 10^{15}\text{ m}$

B $1.496 \times 10^{11}\text{ m}$

C $3.08 \times 10^{16}\text{ m}$

D $3.84 \times 10^8\text{ m}$

Correct Answer: (B)

Core Memory Fact: B** (149.6 million km). **Logic:** (149.6 million km).

Q12 UPPCS PRELIMS 2018

The SI unit of magnetic field strength (H) is:

A Tesla
B Weber
C Ampere per metre
D Henry

Correct Answer: (C)
Core Memory Fact: C** (A/m. Tesla is flux density B). **Logic:** (A/m. Tesla is flux density B).

Q13 UPPCS (PRE) 2017

Which instrument is used to determine the angle of dip of Earth's magnetic field?

A Sextant
B Dip Circle
C Spherometer
D Theodolite

Correct Answer: (B)
Core Memory Fact: B** (Dip Circle). **Logic:** (Dip Circle).

Q14 UPPCS (PRE) 2016

In the human body, the electrical recording of brain activity is denoted by:

A ECG
B EEG
C EMG
D ERG

Correct Answer: (B)
Core Memory Fact: B** (Electroencephalogram). **Logic:** (Electroencephalogram).

Q15 UP RO/ARO 2014

A pressure of 1 bar in CGS units is:

A 10^5 dyne/cm^2
B 10^6 dyne/cm^2
C 10^7 dyne/cm^2
D 10^3 dyne/cm^2

Correct Answer: (B)
Core Memory Fact: B** ($1\text{ bar} = 10^5\text{ N/m}^2 = 10^5 \times 10^5\text{ dyne} / 10^4\text{ cm}^2 = 10^6\text{ dyne/cm}^2$). **Logic:** ($1\text{ bar} = 10^5\text{ N/m}^2 = 10^5 \times 10^5\text{ dyne} / 10^4\text{ cm}^2 = 10^6\text{ dyne/cm}^2$).

Chapter 10: Mechanics, Gravitation and Physical Properties of Matter — UPPCS & UP RO/ARO Drill

15 Authentic Questions • Direct Verified Answers • Active Recall Memory Locks

🎯 High-Yield PYQ Drill (UPPCS & UP RO/ARO)

Exam Drill Rule: High-yield authentic UPPCS & UP RO/ARO questions. Attempt actively; verify against the prominent Memory Lock box below each question.

Q1 UPPCS PRELIMS 2025

Why does an iron needle float horizontally on water when placed carefully, but sink when pushed under?

- **A** Archimedes buoyant force
- **B** Surface tension of the water surface film
- **C** Viscous upthrust
- **D** Atmospheric downward pressure

Correct Answer: (B)

Core Memory Fact: B** (Surface tension forms an elastic skin; once submerged, density $7.8 > 1.0$ causes sinking). **Logic:** (Surface tension forms an elastic skin; once submerged, density $7.8 > 1.0$ causes sinking).

Q2 UPPCS PRELIMS 2024

The Bulk Modulus of an ideal incompressible fluid is:

- **A** Zero
- **B** Unity
- **C** Infinite (∞)
- **D** Negative

Correct Answer: (C)

Core Memory Fact: C** (Incompressible $\implies \Delta V = 0 \implies K = \infty$). **Logic:** (Incompressible $\implies \Delta V = 0 \implies K = \infty$).

Q3 UP RO/ARO PRELIMS 2023

When a drop of oil is placed on clean water, it spreads into a thin film because:

- **A** Oil has higher surface tension than water
- **B** Water has higher surface tension than oil
- **C** Oil is denser than water
- **D** Viscosity of oil is lower than water

Correct Answer: (B)

Core Memory Fact: B** (Water's higher surface tension pulls the oil film outward). **Logic:** (Water's higher surface tension pulls the oil film outward).

Q4 UPPCS PRELIMS 2023

In a capillary tube of radius r, water rises to height h. If the tube radius is halved, water rises to:

A $h/2$
B h
C $2h$
D $4h$

Correct Answer: (C)
Core Memory Fact: C** (Jurin's Law: $h \propto 1/r$). **Logic:** (Jurin's Law: $h \propto 1/r$).

Q5 UPPCS PRELIMS 2022

Which property of a solid determines the extension and behavior of a helical spring under load?

A Young's Modulus
B Bulk Modulus
C Shear Modulus (Modulus of Rigidity)
D Poisson's Ratio

Correct Answer: (C)
Core Memory Fact: Logic:** Recall the Must-Score fact for this stem; eliminate unit/concept twin traps.. **Ans:** **C**.

Q6 UP RO/ARO PRELIMS 2021

The excess pressure inside a liquid drop of radius R and surface tension T is:

A T/R
B $2T/R$
C $4T/R$
D $8T/R$

Correct Answer: (B)
Core Memory Fact: B** (A liquid drop has only one surface, so $\Delta P = 2T/R$; a soap bubble has two surfaces, so $\Delta P = 4T/R$). **Logic:** (A liquid drop has only one surface, so $\Delta P = 2T/R$; a soap bubble has two surfaces, so $\Delta P = 4T/R$).

Q7 UPPCS PRELIMS 2021

As temperature increases, what happens to the viscosity of water and air respectively?

A Both increase
B Both decrease
C Water viscosity decreases, air viscosity increases
D Water viscosity increases, air viscosity decreases

Correct Answer: (C)
Core Memory Fact: Logic:** Recall the Must-Score fact for this stem; eliminate unit/concept twin traps.. **Ans:** **C**.

Q8 UPPCS PRELIMS 2020

A piece of pure ice floats in a glass of water. When the ice melts completely, the water level:

- **A** Rises
- **B** Falls
- **C** Remains unchanged
- **D** First falls then rises

Correct Answer: (C)
Core Memory Fact: Logic:** Recall the Must-Score fact for this stem; eliminate unit/concept twin traps.. **Ans:** **C**.

Q9 UPPCS (MAINS) GS-1

The dimension of the coefficient of linear expansion (α) is:

- **A** $[M^0 L^1 T^0 K^{-1}]$
- **B** $[M^0 L^0 T^0 K^{-1}]$
- **C** $[M^1 L^0 T^0 K^{-1}]$
- **D** $[M^0 L^0 T^{-1} K^0]$

Correct Answer: (B)
Core Memory Fact: B** (K^{-1}).

Q10 UPPCS PRELIMS 2019

A ship floats higher when it travels from a freshwater river into the sea because:

- **A** Sea water is colder
- **B** Sea water is deeper
- **C** Sea water is denser due to dissolved salts
- **D** River water exerts greater viscous drag

Correct Answer: (C)
Core Memory Fact: Logic:** Recall the Must-Score fact for this stem; eliminate unit/concept twin traps.. **Ans:** **C**.

Q11 UP RO/ARO 2016

At what temperature is the surface tension of a liquid reduced to zero?

- **A** Freezing point
- **B** Boiling point
- **C** Absolute zero
- **D** Critical temperature

Correct Answer: (D)
Core Memory Fact: Logic:** Recall the Must-Score fact for this stem; eliminate unit/concept twin traps.. **Ans:** **D**.

Q12 UPPCS PRELIMS 2018

When positive charge is placed on a soap bubble, its radius:

A Decreases
B Increases
C Remains identical
D Flattens into a disc

Correct Answer: (B)
Core Memory Fact: B** (Electrostatic repulsion expands the surface). **Logic:** (Electrostatic repulsion expands the surface).

Q13 UPPCS (PRE) 2017

Torricelli's law states that the velocity of efflux from a tank orifice is proportional to:

A Depth h
B Square root of depth $\sqrt{h}$
C h^2
D Independent of h

Correct Answer: (B)
Core Memory Fact: B** ($v = \sqrt{2gh}$). **Logic:** ($v = \sqrt{2gh}$).

Q14 UPPCS (PRE) 2016

Which of the following is NOT an application of capillarity?

A Kerosene rising in a lamp wick
B Blotting paper absorbing ink
C Drinking coconut water through a straw
D Towel soaking water droplets from the skin

Correct Answer: (C)
Core Memory Fact: C** (Drinking through a straw requires mouth suction pressure reduction). **Logic:** (Drinking through a straw requires mouth suction pressure reduction).

Q15 UP RO/ARO 2014

The spreading parameter S for liquid A on liquid B is positive ($S > 0$). This implies:

A Liquid A beads into droplets
B Liquid A spreads completely over liquid B
C Liquid A sinks to the bottom
D Both liquids boil instantly

Correct Answer: (B)
Core Memory Fact: Logic:** Recall the Must-Score fact for this stem; eliminate unit/concept twin traps.. **Ans:** **B**.

Chapter 11: Energy, Heat and Thermal Science — UPPCS & UP RO/ARO Drill

15 Authentic Questions • Direct Verified Answers • Active Recall Memory Locks

🎯 High-Yield PYQ Drill (UPPCS & UP RO/ARO)

Exam Drill Rule: High-yield authentic UPPCS & UP RO/ARO questions. Attempt actively; verify against the prominent Memory Lock box below each question.

Q1 UPPCS PRELIMS 2025

With reference to heat transfer, which of the following statements is/are correct?
1. Convection is possible in both liquids and gases.
2. Heat can travel through vacuum only by radiation.
3. Conduction involves the bulk physical displacement of heated matter.

Select the correct answer using the code given below:
- (A) 1 and 2 only
- (B) 2 and 3 only
- (C) 1 only
- (D) 1, 2 and 3

Correct Answer: (A)
Core Memory Fact: Logic:** (A) 1 and 2 only **Ans: A.** 1 and 2 only

Q2 UPPCS PRELIMS 2024

Given below are two statements, one labelled as Assertion (A) and the other as Reason (R):
- **Assertion (A):** In severe winters, freshwater lakes freeze from the top surface downwards, leaving liquid water at the bottom.
- **Reason (R):** Water has its maximum density at 4°C due to anomalous thermal expansion.

Select the correct answer:
- (A) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)
- (C) (A) is true but (R) is false
- (D) (A) is false but (R) is true

Correct Answer: (A)
Core Memory Fact: Logic:** (A) Both (A) and (R) are true and (R) is the correct explanation of (A) **Ans: A.** Both (A) and (R) are true and (R) is the correct explanation of (A)

Q3 UP RO/ARO PRELIMS 2023

Match List-I (Thermodynamic Law / Principle) with List-II (Key Scientific Definition):

List-I (Thermodynamic Law / Principle)	List-II (Key Scientific Definition)
A Zeroth Law of Thermodynamics	1 Thermal equilibrium & definition of temperature
B First Law of Thermodynamics	2 Conservation of energy (ΔU = Q - W)
C Second Law of Thermodynamics	3 Entropy always increases in irreversible processes
D Third Law of Thermodynamics	4 Entropy approaches constant minimum at absolute zero

(A) A-1, B-2, C-3, D-4
(B) A-2, B-1, C-4, D-3
(C) A-1, B-3, C-2, D-4
(D) A-4, B-3, C-2, D-1

Correct Answer: (A)
Core Memory Fact: Logic:** (A) A-2, B-1, C-4, D-3 **Ans: A.** A-2, B-1, C-4, D-3

Q4 UPPCS PRELIMS 2023

Which of the following statements is/are correct regarding the specific heat capacity of water?
1. Water's high specific heat capacity makes it an ideal coolant for automotive radiators.
2. It moderates climate swings in coastal geographical zones.
3. Specific heat capacity of liquid water is lower than that of ice.

Select the correct answer using the code given below:
- **(A) 1 and 2 only**
- **(B) 2 and 3 only**
- **(C) 1 only**
- **(D) 1, 2 and 3**

Correct Answer: (A)
Core Memory Fact: Logic:** (A) 1 and 2 only **Ans: A.** 1 and 2 only

Q5 UPPCS PRELIMS 2022

Which of the following pairs is **NOT correctly matched?**
- **(A) Hydrogen — Highest calorific value among fuels**
- **(B) Bimetallic strip — Differential linear thermal expansion**
- **(C) Pressure cooker — Decreases boiling point of water**
- **(D) Thermos flask — Prevents conduction, convection, and radiation**

Correct Answer: (C)
Core Memory Fact: Logic:** (C) Pressure cooker — Decreases boiling point of water **Ans: C.** Pressure cooker — Decreases boiling point of water

Q6 UP RO/ARO PRELIMS 2021

Arrange the following substances in descending order of their thermal conductivities:
1. Silver
2. Glass
3. Aluminum
4. Copper

Select the correct code:
- (A) $1 > 4 > 3 > 2$
- (B) $4 > 1 > 3 > 2$
- (C) $1 > 3 > 4 > 2$
- (D) $3 > 1 > 4 > 2$

Correct Answer: (A)
Core Memory Fact: Logic:** (A) $1 > 4 > 3 > 2$ **Ans: A.** $1 > 4 > 3 > 2$

Q7 UPPCS PRELIMS 2021

With reference to latent heat, which of the following statements is/are correct?
1. Latent heat of vaporization of water is approximately 540 cal/g.
2. Latent heat of fusion of ice is approximately 80 cal/g.
3. Temperature of a substance rises rapidly while absorbing latent heat.

Select the correct answer using the code given below:
- (A) 1 and 2 only
- (B) 2 and 3 only
- (C) 1 only
- (D) 1, 2 and 3

Correct Answer: (A)
Core Memory Fact: Logic:** (A) 1 and 2 only **Ans: A.** 1 and 2 only

Q8 UPPCS PRELIMS 2020

Given below are two statements, one labelled as Assertion (A) and the other as Reason (R):
- **Assertion (A):** In summer, a pendulum clock loses time and runs slow.
- **Reason (R):** The length of the pendulum rod increases due to thermal expansion, increasing its time period of oscillation.

Select the correct answer:
- (A) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)
- (C) (A) is true but (R) is false
- (D) (A) is false but (R) is true

Correct Answer: (A)
Core Memory Fact: Logic:** (A) Both (A) and (R) are true and (R) is the correct explanation of (A) **Ans: A.** Both (A) and (R) are true and (R) is the correct explanation of (A)

Q9 UPPCS (MAINS) GS-1

With reference to radiation laws, which of the following statements is/are correct?
1. Stefan-Boltzmann Law states that radiant power is proportional to the square of absolute temperature.
2. Wien's Law explains why hotter stars radiate maximum power at shorter wavelengths.
3. According to Kirchhoff's Law, good absorbers are poor emitters.

Select the correct answer using the code given below:
- (A) 2 only
- (B) 1 and 2 only
- (C) 2 and 3 only
- (D) 1, 2 and 3

Correct Answer: (A)
Core Memory Fact: Logic:** Standard key matches 2 only. **Ans: A.** 2 only

Q10 UPPCS PRELIMS 2019

Which of the following conditions is **NOT** favorable for the formation of dew?
- (A) Clear sky
- (B) Calm, still air
- (C) High relative humidity
- (D) Heavy overcast cloud cover

Correct Answer: (D)
Core Memory Fact: Logic:** (D) Heavy overcast cloud cover **Ans: D.** Heavy overcast cloud cover

Q11 UP RO/ARO 2016

Match List-I (Everyday Phenomenon) with List-II (Underlying Physics Principle):

List-I		List-II	
A	Cooling of water in an earthen pot	1	Thermal shock from poor thermal conduction
B	Cracking of thick glass with boiling water	2	Evaporative cooling extracting latent heat
C	Operation of a domestic bimetallic thermostat	3	Trapped stationary air layer
D	Heat retention by two thin blankets	4	Differential linear thermal expansion

Select the correct code:
- (A) A-2, B-1, C-4, D-3
- (B) A-1, B-2, C-4, D-3
- (C) A-2, B-4, C-1, D-3
- (D) A-3, B-1, C-4, D-2

Correct Answer: (A)
Core Memory Fact: Logic:** (A) A-2, B-1, C-4, D-3 **Ans: A.** A-2, B-1, C-4, D-3

Q12 UPPCS PRELIMS 2018

Which of the following statements regarding the Carnot engine is correct?
- (A) A Carnot engine can achieve 100% efficiency if operating with an ideal gas
- (B) Efficiency of a Carnot engine depends on the nature of the working fluid
- (C) Efficiency of a Carnot engine depends solely on source and sink temperatures
- (D) Carnot cycle consists of two isobaric and two isochoric processes

Correct Answer: (C)

Core Memory Fact: Logic:** (C) Efficiency of a Carnot engine depends solely on source and sink temperatures **Ans: C.** Efficiency of a Carnot engine depends solely on source and sink temperatures

Q13 UPPCS (PRE) 2017

With reference to thermal expansion of water, which of the following statements is/are correct?
1. When water is heated from 0°C to 4°C, its volume expands continuously.
2. At 4°C, density of water is at its maximum value.
3. Above 4°C, water expands uniformly like normal liquids.

Select the correct answer using the code given below:
- (A) 2 and 3 only
- (B) 1 and 2 only
- (C) 1 only
- (D) 1, 2 and 3

Correct Answer: (A)

Core Memory Fact: Logic:** (A) 2 and 3 only **Ans: A.** 2 and 3 only

Q14 UPPCS (PRE) 2016

Given below are two statements, one labelled as Assertion (A) and the other as Reason (R):
- **Assertion (A):** Running a domestic refrigerator with its door open inside a closed room heats up the room.
- **Reason (R):** The heat rejected by the condenser coils into the room exceeds the heat extracted from the interior by the amount of electrical work performed by the compressor motor.

Select the correct answer:
- (A) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)
- (C) (A) is true but (R) is false
- (D) (A) is false but (R) is true

Correct Answer: (A)

Core Memory Fact: Logic:** (A) Both (A) and (R) are true and (R) is the correct explanation of (A) **Ans: A.** Both (A) and (R) are true and (R) is the correct explanation of (A)

Q15 UP RO/ARO 2014

Which of the following units represents energy?
1. Kilowatt-hour (kWh)
2. Electron-volt (eV)
3. Joule
4. Watt

Select the correct answer using the code given below:
- **(A) 1, 2 and 3 only**
- **(B) 1 and 3 only**
- **(C) 3 and 4 only**
- **(D) 1, 2, 3 and 4**

Correct Answer: (A)
Core Memory Fact: Logic:** (A) 1, 2 and 3 only **Ans: A.** 1, 2 and 3 only

Chapter 12: Light, Optics and Laser Technology — UPPCS & UP RO/ARO Drill

15 Authentic Questions • Direct Verified Answers • Active Recall Memory Locks

🎯 High-Yield PYQ Drill (UPPCS & UP RO/ARO)

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Q1 UPPCS PRELIMS 2025

What is the focal length of a flat plane mirror?

A Zero

B $+1\text{ metre}$

C -1 metre

D Infinite (∞)

Correct Answer: (D)

Core Memory Fact: Logic:** Recall the Must-Score fact for this stem; eliminate unit/concept twin traps.. **Ans:** **D**.

Q2 UPPCS PRELIMS 2024

An air bubble trapped inside clear water behaves optically as a:

A Concave mirror

B Convex lens

C Concave (divergent) lens

D Cylindrical lens

Correct Answer: (C)

Core Memory Fact: C** ($\mu_{\text{water}} > \mu_{\text{air}}$). **Logic:** ($\mu_{\text{water}} > \mu_{\text{air}}$).

Q3 UP RO/ARO PRELIMS 2023

A person standing 2 metres in front of a plane mirror moves 0.5 metres closer to it. The distance between the person and their image is:

A 1.5 m

B 3.0 m

C 4.0 m

D 1.0 m

Correct Answer: (B)

Core Memory Fact: B** (Object distance = 1.5 m; image distance = 1.5 m; total distance = 3.0 m). **Logic:** (Object distance = 1.5 m; image distance = 1.5 m; total distance = 3.0 m).

Q4 UPPCS PRELIMS 2023

Which physical phenomenon is primarily responsible for the formation of shadows?

A Refraction of light
B Rectilinear propagation of light
C Diffraction of light
D Polarisation of light

Correct Answer: (B)
Core Memory Fact: Logic:** Recall the Must-Score fact for this stem; eliminate unit/concept twin traps.. **Ans:** **B**.

Q5 UPPCS PRELIMS 2022

In a color television screen, pixels emit which primary colors?

A Red, Yellow, Blue
B Red, Green, Blue
C Cyan, Magenta, Yellow
D Red, Green, Yellow

Correct Answer: (B)
Core Memory Fact: B** (Additive RGB phosphors). **Logic:** (Additive RGB phosphors).

Q6 UP RO/ARO PRELIMS 2021

The power of a reading glass lens with a focal length of $+25\text{ cm}$ is:

A $+1.0\text{ D}$
B $+2.5\text{ D}$
C $+4.0\text{ D}$
D $+0.25\text{ D}$

Correct Answer: (C)
Core Memory Fact: C** ($P = 100/25 = +4.0\text{ Dioptres}$). **Logic:** ($P = 100/25 = +4.0\text{ Dioptres}$).

Q7 UPPCS PRELIMS 2021

The sky appears blue from Earth, but to an astronaut on the International Space Station, it appears:

A White
B Blue
C Black
D Violet

Correct Answer: (C)
Core Memory Fact: C** (Absence of atmospheric scattering). **Logic:** (Absence of atmospheric scattering).

Q8 UPPCS PRELIMS 2020

A doctor examining a patient's throat or teeth uses a:

A Convex lens
B Convex mirror
C Concave mirror
D Cylindrical mirror

Correct Answer: (C)
Core Memory Fact: C** (Produces erect magnified virtual image up close). **Logic:** (Produces erect magnified virtual image up close).

Q9 UPPCS (MAINS) GS-1

If a plane mirror is turned through an angle of 15°, the reflected light beam turns through:

A 15°
B 30°
C 45°
D 0°

Correct Answer: (B)
Core Memory Fact: B** ($2\theta = 2 \times 15^\circ = 30^\circ$). **Logic:** ($2\theta = 2 \times 15^\circ = 30^\circ$).

Q10 UPPCS PRELIMS 2019

A rainbow is never visible in the sky at:

A 08:00 AM
B 12:00 Noon
C 04:00 PM
D 06:00 PM

Correct Answer: (B)
Core Memory Fact: B** (Sun is overhead; rainbow forms opposite the Sun). **Logic:** (Sun is overhead; rainbow forms opposite the Sun).

Q11 UP RO/ARO 2016

The image formed on the retina of the human eye is:

A Real, inverted, and magnified
B Virtual, erect, and magnified
C Real, inverted, and diminished
D Virtual, erect, and diminished

Correct Answer: (C)
Core Memory Fact: Logic:** Recall the Must-Score fact for this stem; eliminate unit/concept twin traps.. **Ans:** **C**.

Q12 UPPCS PRELIMS 2018

In optical fibers, light is transmitted along the glass core via:

A Refraction
B Total internal reflection
C Light scattering
D Diffraction

Correct Answer: (B)
Core Memory Fact: Logic:** Recall the Must-Score fact for this stem; eliminate unit/concept twin traps.. **Ans:** **B**.

Q13 UPPCS (PRE) 2017

Which color of light experiences the maximum deviation when white light passes through a glass prism?

A Red
B Yellow
C Green
D Violet

Correct Answer: (D)
Core Memory Fact: Logic:** Recall the Must-Score fact for this stem; eliminate unit/concept twin traps.. **Ans:** **D**.

Q14 UPPCS (PRE) 2016

The minimum height of a vertical plane mirror in which a 1.8 m tall person can view their entire reflection is:

A 0.6 m
B 0.9 m
C 1.8 m
D 3.6 m

Correct Answer: (B)
Core Memory Fact: B** ($1.8 / 2 = 0.9\text{ m}$). **Logic:** ($1.8 / 2 = 0.9\text{ m}$).

Q15 UP RO/ARO 2014

A cut diamond sparkles brilliantly because of its:

A Large critical angle
B High electrical conductivity
C Very small critical angle and total internal reflection
D Absorption of infrared rays

Correct Answer: (C)
Core Memory Fact: C** ($i_c \approx 24.4^\circ$). **Logic:** ($i_c \approx 24.4^\circ$).

Chapter 13: Sound and Wave Motion — UPPCS & UP RO/ARO Drill

15 Authentic Questions • Direct Verified Answers • Active Recall Memory Locks

🎯 High-Yield PYQ Drill (UPPCS & UP RO/ARO)

Exam Drill Rule: High-yield authentic UPPCS & UP RO/ARO questions. Attempt actively; verify against the prominent Memory Lock box below each question.

Q1 UPPCS PRELIMS 2025

A sound wave travels from water into steel. Which of the following quantities remains completely unchanged?
- (A) Velocity
- (B) Wavelength
- (C) Frequency
- (D) Amplitude

Correct Answer: (C)
Core Memory Fact: Logic:** Frequency is a fundamental characteristic of the vibrating source. **Ans: C.** Frequency

Q2 UPPCS PRELIMS 2024

An astronaut on the Moon's surface drops a heavy hammer and a feather from the same height. If an explosion occurs concurrently 500 m away:
- (A) He hears the explosion before seeing the hammer hit the surface
- (B) He hears the explosion after seeing the hammer hit the surface
- (C) He cannot hear the explosion at all
- (D) The sound travels faster on the Moon due to low gravity

Correct Answer: (C)
Core Memory Fact: Logic:** The Moon has no atmosphere (vacuum). **Ans: C.** He cannot hear the explosion at all

Q3 UP RO/ARO PRELIMS 2023

What is the minimum distance required between an observer and a cliff to hear a clear echo at 0°C (Speed of sound in air at $0^\circ\text{C} = 331\text{ m/s}$)?
- (A) 17.2 m
- (B) 16.55 m
- (C) 33.1 m
- (D) 8.28 m

Correct Answer: (B)
Core Memory Fact: Logic:** Persistence of hearing $= 0.1\text{ s}$. **Ans: B.** 16.55 m

Q4 UPPCS PRELIMS 2023

The technique of crushing kidney stones using high-intensity sound waves without surgical incision is termed:
- (A) Endoscopy
- (B) Lithotripsy
- (C) Dialysis
- (D) Ultrasonography

Correct Answer: (B)

Core Memory Fact: Logic:** Extracorporeal Shock Wave Lithotripsy (ESWL) uses focused ultrasonic shockwaves to break kidney stones into small particles excreted via urine. **Ans: B.** Lithotripsy

Q5 UPPCS PRELIMS 2022

Which of the following statements regarding FM (Frequency Modulation) is NOT correct?
- (A) It was invented by Edwin H. Armstrong
- (B) It has wider bandwidth than Amplitude Modulation
- (C) It is less susceptible to atmospheric noise and static
- (D) Carrier amplitude continuously changes according to the signal

Correct Answer: (D)

Core Memory Fact: Logic:** In FM, the amplitude and phase of the carrier wave remain strictly constant; only its frequency varies in accordance with the modulating audio signal. **Ans: D.** Carrier amplitude continuously changes according to the signal

Q6 (UP RO/ARO (PRE) 2021)

In a Sitar, which type of sound vibrations are produced?

(a) Progressive and Longitudinal

(b) Progressive and Transverse

(c) Stationary and Longitudinal

(d) Stationary and Transverse

Correct Answer: (D)

Core Memory Fact: Logic:** When we pluck the string of an instrument, like the Sitar, the sound that we hear is not only that of the string. **Ans: D.** Stationary and Transverse

Q7 (UPPCS (PRE) 1991)

Long radio waves are reflected by which of the

following layer of Earth's surface –

(a) Troposphere (b) Ionosphere

(c) Tropopause (d) Stratosphere

Correct Answer: (B)

Core Memory Fact: Logic:** The ionosphere is a region of Earth's upper atmosphere from about 50 to 400 miles altitude. **Ans: B.** Ionosphere

Q8 (UPPCS (PRE) 1998)

Wireless communication is reflected back to the earth's

surface by the –

(a) Troposphere (b) Stratosphere

(c) Ionosphere (d) Exosphere

Correct Answer: (C)
Core Memory Fact: Logic:** Standard key matches Ionosphere (d) Exosphere. **Ans: C.** Ionosphere (d) Exosphere

Q9 UP RO/ARO PRELIMS 2021

Which of the following atmospheric layers is

responsible for the deflection of radiowaves ?

(a) Troposphere (b) Stratosphere

(c) Mesosphere (d) Ionosphere

Correct Answer: (D)
Core Memory Fact: Ionosphere

Q10 (UPPCS (PRE) 2005)

Which one of the following statements is not true about

cosmic rays?

(a) They are electromagnetic waves.

(b) They have very short wavelength.

(c) They are made of highly energetic charged particles.

(d) They originate from the Sun

Correct Answer: (A)
Core Memory Fact: Logic:** Standard key matches They are electromagnetic waves. **Ans: A.** They are electromagnetic waves

Q11 (UP RO/ARO (MAINS) 2017)

Which of the following does not require a medium?

(a) Radiation (b) Convection

(c) Conduction (d) None of these

Correct Answer: (A)

Core Memory Fact: Logic:** In physics, radiation is the emission or transmission of energy in the form of waves or particles through space or through a medium. **Ans: A.** Radiation (b) Convection

Q12 (UPPCS (PRE) 2019)

Which one of the following is associated with 'Albedo'?

(a) Transmitting power (b) Absorbing power

(c) Emissive power (d) Reflecting power

Correct Answer: (D)

Core Memory Fact: Logic:** Albedo is the fraction of light or radiation that is reflected by a body or surface. **Ans: D.** Reflecting power

Q13 (UPPCS (PRE) 2022)

The Earth's atmosphere is mainly heated by which one

of the following?

(a) Scattered solar radiation

(b) Shortwave solar radiation

(c) Longwave terrestrial radiation

(d) Reflected solar radiation

Correct Answer: (C)

Core Memory Fact: Logic:** The energy released from the Sun is emitted as shortwave light and ultraviolet energy. **Ans: C.** Longwave terrestrial radiation

Q14 (UPPCS (PRE) 2018 / UPPCS (MAINS) 2005)

Which of the following electromagnetic radiations has

the maximum energy?

(a) Visible light (b) Infrared rays

(c) Ultraviolet rays (d) X-rays

Correct Answer: (D)

Core Memory Fact: Logic:** Shorter the wavelength, greater is the frequency and energy. **Ans: D.** X-rays

Q15 (UP RO/ARO (PRE) 2023)

Consider the following and arrange these in increasing

order of frequency :

I. X-rays II. Visible rays

III. Infrared rays IV. Radio waves

Select the correct answer using the code given below :

Code :

(a) III, II, I, IV (b) I, II, III, IV

(c) II, III, IV, I (d) IV, III, II, I

Correct Answer: (D)

Core Memory Fact: Logic:** The correct increasing order of the given electromagnetic waves in terms of frequency is as follows : Radio waves < Infrared rays < Visible rays < X-rays. **Ans: D.** IV, III, II, I

Chapter 14: Electricity, Magnetism and Electromagnetism — UPPCS & UP RO/ARO Drill

15 Authentic Questions • Direct Verified Answers • Active Recall Memory Locks

🎯 High-Yield PYQ Drill (UPPCS & UP RO/ARO)

Exam Drill Rule: High-yield authentic UPPCS & UP RO/ARO questions. Attempt actively; verify against the prominent Memory Lock box below each question.

Q1 UPPCS PRELIMS 2025

What is the total electrical energy consumed by a household using three 100 W fans for 10 hours each day in a month of 30 days?
- (A) 9 units
- (B) 30 units
- (C) 90 units
- (D) 900 units

Correct Answer: (C)

Core Memory Fact: Logic:** $\text{Units} = \frac{3 \times 100\text{ W} \times 10\text{ h/day} \times 30\text{ days}}{1000} = 90\text{ kWh (units)}$. **Ans: C.** 90 units

Q2 UPPCS PRELIMS 2024

Two electric bulbs rated $25\text{ W}, 220\text{ V}$ and $100\text{ W}, 220\text{ V}$ are connected in series across a 220 V supply. Which bulb will glow brighter?
- (A) 100 W bulb
- (B) 25 W bulb
- (C) Both will glow with equal brightness
- (D) Neither will glow

Correct Answer: (B)

Core Memory Fact: Logic:** $R = V^2 / P$. **Ans: B.** 25 W bulb

Q3 UP RO/ARO PRELIMS 2023

An electric fuse wire melts at 5 A. If a fuse of the same material is to be made with a current rating of 10 A, the new wire must have:
- (A) Smaller radius
- (B) Larger radius
- (C) Greater length
- (D) Smaller length

Correct Answer: (B)

Core Memory Fact: Logic:** By Preece's Law for fuse wires, the fusing current $I \propto r^{3/2}$. **Ans: B.** Larger radius

Q4 UPPCS PRELIMS 2023

A transformer is used to:
- (A) Convert AC into DC
- (B) Convert DC into AC
- (C) Step-up or step-down AC voltage
- (D) Step-up or step-down DC voltage

Correct Answer: (C)

Core Memory Fact: Logic:** Transformers operate strictly on alternating current via electromagnetic induction; they cannot function on direct current. **Ans: C.** Step-up or step-down AC voltage

Q5 UPPCS PRELIMS 2022

In an electric vehicle Lithium-ion battery, which of the following materials is typically utilized for the ANODE?
- (A) Cobalt oxide
- (B) Lithium phosphate
- (C) Nickel oxide
- (D) Graphite

Correct Answer: (D)

Core Memory Fact: Logic:** The anode is made of graphite (carbon) coated on copper foil, which intercalates lithium ions during charging. **Ans: D.** Graphite

Q6 (UP RO/ARO (PRE) 2023)

15 bulbs each of 200 watt capacity in a Silkyara

tunnel are illuminated continuously for 24 hours. The

consumption of electricity will be :

(a) 71 units (b) 72 units

(c) 73 units (d) None of the above

Correct Answer: (B)

Core Memory Fact: Logic:** Consumption of electricity by one bulb = watt × hours 1000 = 200 × 24 1000 = 4.8 units Hence, total consumption of electricity by 15 bulbs = 15 × 4.8 = 72 units **Ans: B.** 72 units

Q7 (UPPCS (PRE) 2017)

Five bulbs each of 100 watt capacity in a tunnel

are illuminated continuously for twenty hours. The

consumption of electricity will be:

(a) One unit (b) Two units

(c) Ten units (d) Twenty units

Correct Answer: (C)

Core Memory Fact: Logic:** Consumption of electricity by one bulb = 2 units Thus, consumption of electricity by 5 bulbs = 2 × 5 = 10 units **Ans: C.** Ten units (d) Twenty units

Q8 (UPPCS (PRE) 2006)

The value of 1 kilowatt hour is –

(a) 3.6 × 106

J (b) 3.6 × 103

J

(c) 103 J (d) 105

J

Correct Answer: (A)

Core Memory Fact: Logic:** 1 Kilowatt hour = 1 (kilowatt) (1 hour) = (1000 joule/sec.) (3600 sec.) = 36 × 105 Joule = 3.6 × 106 Joule Thus, option (a) is correct. **Ans: A.** 3.6 × 106

Q9 UP RO/ARO PRELIMS 2021

An electric bulb of 100 watt is used for 4 hours. The

unit of electric energy used is –

(a) 400 (b) 25

(c) 4 (d) 0.4

Correct Answer: (D)

Core Memory Fact: Logic:** The electric energy consumed = 100 watt × 4 hour = 400 watt hour kilowatt hour = 0.4 kilowatt hour = 0.4 unit **Ans: D.** 0.4

Q10 (UP LOWER SUB (PRE) 2008 / UPPCS (PRE) 2009)

Of the two bulbs in a house, one glows brighter than the other. In this context which of the following statements is correct ?

(a) The brightness does not depend on the resistance.

(b) Both the bulbs have same the resistance.

(c) The bright bulb has larger resistance.

(d) The dim bulb has larger resistance.

Correct Answer: (D)

Core Memory Fact: Logic:** The bulb which decays more energy will generate more brightness. **Ans: D.** The dim bulb has larger resistance

Q11 (UPPCS (MAINS) 2013)

Of the two bulbs in a house, one glows brighter than the other. Which of the two has a larger resistance ?

(a) The dim bulb

(b) The brighter bulb

(c) Both have the same resistance

(d) The brightness does not depend on the resistance

Correct Answer: (A)

Core Memory Fact: Logic:** Standard key matches The dim bulb. **Ans: A.** The dim bulb

Q12 (UP RO/ARO (PRE) 2023)

A voltage of 15 volt is applied across a colour coded

carbon resistor with first, second and third rings of

brown, green and black colours. What is the current

flowing through the resistor?

(a) 3 ampere (b) 2 ampere

(c) 1 ampere (d) None of the above

Correct Answer: (C)

Core Memory Fact: Logic:** Resistor code value for brown colour = 1 (A) Resistor code value for green colour = 5 (B) Resistor code value for black colour = 0 (C) We know that first two bands (rings) indicate the first two significant digits of re... **Ans: C.** 1 ampere (d) None of the above

Q13 (UPPCS (MAINS) 2009)

A dynamo which is said to generate electricity actually

acts as a :

(a) Source of ions

(b) Source of electric charge

(c) Converter of energy

(d) Source of electrons

Correct Answer: (A)

Core Memory Fact: Logic:** A dynamo, which is also known as an electrical generator produces direct current through a commutator. **Ans: C.** Converter of energy

Q14 (UPPCS (PRE) 2016 / 39TH B.P.S.C. (PRE) 1994 / UPPCS (PRE) 1993 / UPPCS (PRE) 1992)

'Dynamo' is a device, which converts :

(a) Chemical energy into electrical energy

(b) Electrical energy into mechanical energy

(c) Mechanical energy into electrical energy

(d) Electrical energy into chemical energy

Correct Answer: (A)

Core Memory Fact: Logic:** A dynamo is an electrical generator that produces direct current with the use of a commutator. **Ans: C.** Mechanical energy into electrical energy

Q15 (UPPCS (MAINS) 2012 / RAS/RTS(PRE) 2012)

Dynamo is a machine which is used for –

(a) Conversion of high voltage to low voltage

(b) Conversion of electrical energy to mechanical energy

(c) Conversion of mechanical energy to electrical energy

(d) Conversion of low voltage to high voltage

Correct Answer: (C)

Core Memory Fact: Logic:** Standard key matches Conversion of mechanical energy to electrical energy. **Ans: C.** Conversion of mechanical energy to electrical energy

Chapter 15: Electronics, Semiconductors and Computer Technology — UPPCS & UP RO/ARO Drill

15 Authentic Questions • Direct Verified Answers • Active Recall Memory Locks

🎯 High-Yield PYQ Drill (UPPCS & UP RO/ARO)

Exam Drill Rule: High-yield authentic UPPCS & UP RO/ARO questions. Attempt actively; verify against the prominent Memory Lock box below each question.

Q1 UPPCS PRELIMS 2025

What is the minimum length of a standard MAC (Media Access Control) address?
- (A) 16 bits
- (B) 32 bits
- (C) 48 bits
- (D) 64 bits

Correct Answer: (A)

Core Memory Fact: Logic:** A MAC address is a 48-bit (6-byte) physical hardware address burned into the network interface card (NIC), operating at the Data Link Layer (Layer 2) of the OSI model. **Ans: C.** 48 bits

Q2 UPPCS PRELIMS 2024

Which of the following optical disc formats utilizes a blue-violet laser with a wavelength of 405 nm?
- (A) CD-ROM
- (B) DVD-R
- (C) Blu-ray Disc
- (D) Floppy Disk

Correct Answer: (C)

Core Memory Fact: Logic:** Blu-ray Discs use a 405 nm blue-violet laser, allowing a much tighter focus than the 650 nm red laser of DVDs, giving 25 GB storage per layer. **Ans: C.** Blu-ray Disc

Q3 UP RO/ARO PRELIMS 2023

In a 4-bit binary weighted code, which of the following is an example of a self-complementing code?
- (A) 8421 code
- (B) 5211 code
- (C) 5421 code
- (D) Gray code

Correct Answer: (A)

Core Memory Fact: Logic:** A code is self-complementing if the sum of its weights equals 9. **Ans: B.** 5211 code

Q4 UPPCS PRELIMS 2023

The international AI Action Summit held in February 2025 took place in:
- (A) London
- (B) Seoul
- (C) Paris
- (D) New Delhi

Correct Answer: (C)
Core Memory Fact: Logic:** Standard key matches Paris. **Ans: C.** Paris

Q5 UPPCS PRELIMS 2022

In the context of computer security, a software that disguises itself as useful legitimate software while executing unauthorized background actions without self-replicating is called:
- (A) Virus
- (B) Worm
- (C) Trojan Horse
- (D) Keylogger

Correct Answer: (C)
Core Memory Fact: Logic:** Unlike viruses and worms, Trojans do not attempt to self-replicate or inject code into other files. **Ans: C.** Trojan Horse

Q6 (UPPCS (MAINS) 2015)

Which one of the following is the best conductor of

electricity?

(a) Mica (b) Copper

(c) Gold (d) Silver

Correct Answer: (D)
Core Memory Fact: Silver

Q7 (UPPCS (MAINS) 2012 / RAS/RTS(PRE) 2012)

Which of the following is the best conductor of

electricity?

(a) Aluminium (b) Copper

(c) Silver (d) Gold

Correct Answer: (C)
Core Memory Fact: Logic:** Standard key matches Silver (d) Gold. **Ans: C.** Silver (d) Gold

Q8 (UP RO/ARO (MAINS) 2013)

Assertion (A): Copper rods are generally preferred

to iron rods for making lightning

conductors.

Reason (R) : Copper is better conductor of electricity

than iron and is not easily oxidized

under atmospheric conditions.

Code :

(a) Both (A) and (R) are true and (R) is the correct

explanation of (A).

(b) Both (A) and (R) are true, but (R) is not the correct

explanation of (A).

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G–141 General Studies General Science

(c) (A) is true, but (R) is false.

(d) (A) is false, but (R) is true.

Correct Answer: (A)

Core Memory Fact: Logic:** Copper rods are preferred for making lightning conductor because : (i) Copper is a better conductor of electricity than iron. **Ans: A.** Both (A) and (R) are true and (R) is the correct

Q9 (UP RO/ARO (PRE) 2023)

Given below are two statements, in which one is

labelled as Assertion (A) and the other as Reason (R).

Assertion (A) : A person travelling in a car with closed

glass windows and doors is safe from the lightening

strike.

Reason (R) : Actually a closed car behaves like a hollow

conductor, so charges cannot enter inside the car.

Select the correct answer using the code given belowCode :

(a) (A) is true but (R) is false.

(b) Both (A) and (R) are true and (R) is correct explanation

of (A).

(c) (A) is false but (R) is true.

(d) Both (A) and (R) are true but (R) is not correct

explanation of (A).

Correct Answer: (A)

Core Memory Fact: Logic:** A person travelling in a car with closed glass windows and doors is safe from the lightening strike, because closed car behaves like a closed hollow conductor (as a Faraday cage) and electric field or charge inside a cl... **Ans: B.** Both (A) and (R) are true and (R) is correct explanation

Q10 (UPPCS (PRE) 2000)

At which temperature superconductivity can be

of tremendous economic interest saving billions of

rupees?

(a) At very low temperature

(b) At a temperature when semiconductor becomes a

superconductor

(c) at room temperature

(d) At a very high temperature

Correct Answer: (C)

Core Memory Fact: Logic:** If we could achieve superconductivity at room temperature then we can save billions of rupee through electric transmission without any power loss. **Ans: C.** at room temperature

Q11 UP RO/ARO PRELIMS 2021

The highest temperature attained by a superconductor

is :

(a) 24 K (b) 133 K

(c) 150 K (d) 300 K

Correct Answer: (C)

Core Memory Fact: Logic:** When superconductivity was discovered in 1911, it was found only at temperatures close to absolute zero (– 273.15°C). **Ans:** When superconductivity was discovered in 1911, it was found only at temperatures close to absolute zero (– 273.15°C).

Q12 (UTTARAKHAND P.C.S. (PRE) 2006 / UPPCS (PRE) 2000)

The newly discovered high temperature super

conductors are –

(a) Metal alloys (b) Pure rare earth metals

(c) Ceramic oxides (d) Inorganic polymers

Correct Answer: (C)

Core Memory Fact: Logic:** Ceramic oxides play a vital role in the field of research and discovery towards superconductivity. **Ans: C.** Ceramic oxides

Q13 (UPPCS (MAINS) 2015)

The resistance of a semiconductor on heating :

(a) Remains constant (b) Decreases

(c) Increases (d) None of the above

Correct Answer: (B)
Core Memory Fact: Logic:** With the increase in temperature, the conductivity of the semiconductor material increases. **Ans: B.** Decreases

Q14 (UPPCS (PRE) 2015)

The most commonly used material for making

transistors is –

(a) Aluminium (b) Silicon

(c) Copper (d) Silver

Correct Answer: (B)
Core Memory Fact: Silicon

Q15 (UP RO/ARO (PRE) 2016)

Which one of the following is an important component

of a transistor?

(a) Arsenic (b) Germanium

(c) Osmium (d) Radium

Correct Answer: (B)
Core Memory Fact: Germanium

Chapter 16: Nuclear and Atomic Physics — UPPCS & UP RO/ARO Drill

15 Authentic Questions • Direct Verified Answers • Active Recall Memory Locks

🎯 High-Yield PYQ Drill (UPPCS & UP RO/ARO)

Exam Drill Rule: High-yield authentic UPPCS & UP RO/ARO questions. Attempt actively; verify against the prominent Memory Lock box below each question.

Q1 UPPCS PRELIMS 2025

Which of the following is NOT used as a moderator in a nuclear reactor?
- (A) Light water
- (B) Heavy water
- (C) Cadmium
- (D) Graphite

Correct Answer: (C)
Core Memory Fact: Logic:** Standard key matches Cadmium. **Ans: C.** Cadmium

Q2 UPPCS PRELIMS 2024

The Prototype Fast Breeder Reactor (PFBR) with a capacity of 500 MWe in India is situated at:
- (A) Narora, Uttar Pradesh
- (B) Rawatbhata, Rajasthan
- (C) Kalpakkam, Tamil Nadu
- (D) Kaiga, Karnataka

Correct Answer: (C)
Core Memory Fact: Logic:** The 500 MWe PFBR is located at Kalpakkam, Tamil Nadu, forming Stage 2 of India's Three-Stage Nuclear Power Programme. **Ans: C.** Kalpakkam, Tamil Nadu

Q3 UP RO/ARO PRELIMS 2023

An instrument used to detect the presence of gamma-emitting radioactive isotopes in drinking water is:
- (A) Spectrophotometer
- (B) Scintillation Counter
- (C) Compound Microscope
- (D) Barometer

Correct Answer: (B)
Core Memory Fact: Logic:** Scintillation counters detect ionizing radiation by measuring the light flashes produced in scintillator crystals upon gamma-ray absorption. **Ans: B.** Scintillation Counter

Q4 UPPCS PRELIMS 2023

Natural uranium contains what percentage of the fissile isotope Uranium-235?
- (A) Approximately 0.7%
- (B) Approximately 3.5%
- (C) Approximately 20%
- (D) Approximately 90%

Correct Answer: (A)

Core Memory Fact: Logic:** Natural uranium consists of 99.3% U-238 and only 0.7% fissile U-235. **Ans: A.** Approximately 0.7%

Q5 UPPCS PRELIMS 2022

In India, which entity possesses exclusive statutory authorization to process monazite sand and extract thorium?
- (A) Private placer mining consortia
- (B) Indian Rare Earths Limited (IREL)
- (C) National Aluminium Company (NALCO)
- (D) Minerals and Metals Trading Corporation (MMTC)

Correct Answer: (B)

Core Memory Fact: Logic:** Under the Atomic Energy Act, 1962, monazite is a prescribed strategic substance; only government undertakings like IREL (under DAE) and KMML can process monazite. **Ans: B.** Indian Rare Earths Limited (IREL)

Q6 (U.P. U.D.A./L.D.A. (PRE) 2002 / UPPCS (PRE) 2001, 2003)

Which one of the following pairs is not correctly

matched :

(a) Discovery of Meson – Hideki Yukawa

(b) Discovery of – C.D. Anderson

Positron and U.F. Hess

(c) Theory of energy – H.A. Bethe

production in the

sun and stars

(d) Synthesis of – Enrico Fermi

transuranic elements

Correct Answer: (D)

Core Memory Fact: Logic:** In 1935 Meson particles were discovered by Japanese physicist Hideki Yukawa. **Ans: D.** Synthesis of – Enrico Fermi

Q7 (UPPCS (PRE) 1999)

Which one of the following is not correct :

(a) Theory of evolution was propounded by Charles Darwin.

(b) The breaking apart of the nucleus of an atom is called

fusion.

(c) 'Dry ice' is nothing but solid carbon dioxide.

(d) Telephone was invented by Graham Bell.

Correct Answer: (B)

Core Memory Fact: Logic:** Among the given options, option (b) is not correct. **Ans: B.** The breaking apart of the nucleus of an atom is called

Q8 (UPPCS (PRE) 1994)

Read the following statements–

Statement (A): Ernest Rutherford said in Royal Society

that a man will never be able to get

nuclear energy.

Reason (R) : He was confident that the Law of

Einstein will fail and the volume will

not be converted to energy.

Select your answer in the code given below :

(a) Both (A) and (R) are true and (A) is the correct

explanation of (A).

(b) Both (A) and (R) are true but (R) is not the correct

explanation of (A).

(c) (A) is correct but (R) is not correct.

(d) (R) is correct but (A) is not correct.

Correct Answer: (A)

Core Memory Fact: Logic:** Both given statement and reason are correct and reason (R) is the correct explanation of statement (A). **Ans: A.** Both (A) and (R) are true and (A) is the correct

Q9 (UPPCS (PRE) 2009)

One of these particles is claimed to have invented which

rebut the Einstein's theory of relativity :

(a) Microwave photon (b) Neutrino

(c) Liquid crystal (d) Light emitting diode

Correct Answer: (A)
Core Memory Fact: Logic:** The two physicists Gunter Nimtz and Alfons Stahlhofen of the University of Koblenz, Germany claimed that they had propelled microwave photons faster than the speed of light. **Ans: A.** Microwave photon

Q10 (UPPCS (PRE) 1997)

Cyclotrons are used to accelerate :

(a) Neutrons (b) Protons

(c) Atoms (d) Ions

Correct Answer: (B)
Core Memory Fact: Logic:** Cyclotron is a machine used to accelerate charged particle such as alpha particles, deuteron, proton etc. **Ans: B.** Protons

Q11 (UPPCS (PRE) 2009)

Which one of the following is the source of renewable

energy in stars?

(a) Hydrogen changes into helium

(b) Helium changes into hydrogen

(c) Decay of radioactive material

(d) Excess of oxygen which is helpful in burning and

generate energy.

Correct Answer: (A)
Core Memory Fact: Logic:** Standard key matches Hydrogen changes into helium. **Ans: A.** Hydrogen changes into helium

Q12 (UPPCS (PRE) 1992)

What is a Nuclear Reactor?

(a) Place where atomic bomb is built

(b) Heavy water pond

(c) The emitter of U238

(d) Molecular furnace

Correct Answer: (D)
Core Memory Fact: Logic:** Standard key matches Molecular furnace. **Ans: D.** Molecular furnace

Q13 (UP RO/ARO (PRE) 2016)

Consider the following statements about nuclear fusion

reactors :

i. They work on the principle of fission of heavy nuclei.

ii. They usually have Tokamak design.

iii. They operate at very high temperature.

Of these

(a) Only i and iii are correct

(b) Only i and ii are correct

(c) Only ii and iii are correct

(d) All the three i, ii and iii are correct

Correct Answer: (C)
Core Memory Fact: Logic:** Nuclear fusion reactors work on the principle of fusion of two light nuclei. **Ans: C.** Only ii and iii are correct

Q14 (UPPCS (PRE) 1999 / I.A.S. (PRE) 1995)

The difference between nuclear reactor and atom bomb

is that :

(a) There is no chain reaction in nuclear reactor whereas

this happens in atom bomb.

(b) The chain reaction in nuclear reactor is controlled.

(c) The chain reaction in nuclear reactor is not controlled.

(d) Atom bomb is based on nuclear fusion whereas in

nuclear reactor, nuclear fission occurs.

Correct Answer: (B)

Core Memory Fact: Logic:** The chain reaction process used in nuclear reactors and atom bombs both release vast amount of energy, but the manner in which the energy is controlled and utilized is different in both the cases. **Ans: B.** The chain reaction in nuclear reactor is controlled

Q15 UP RO/ARO PRELIMS 2021

The important nuclear fuel available in India in

abundance is :

(a) Uranium (b) Thorium

(c) Iridium (d) Plutonium

Correct Answer: (B)

Core Memory Fact: Logic:** India is one country that has an abundance of an important nuclear fuel thorium. **Ans: B.** Thorium

Chapter 17: Scientists, Discoveries, Defence & Space Technology, and Applications of Physics — UPPCS & UP RO/ARO Drill

15 Authentic Questions • Direct Verified Answers • Active Recall Memory Locks

🎯 High-Yield PYQ Drill (UPPCS & UP RO/ARO)

Exam Drill Rule: High-yield authentic UPPCS & UP RO/ARO questions. Attempt actively; verify against the prominent Memory Lock box below each question.

Q1 UPPCS PRELIMS 2025

1. It was the first flight test of the Agni-V missile equipped with Multiple Independently Targetable Re-entry Vehicle (MIRV) technology.
2. The test was conducted from Dr. APJ Abdul Kalam Island off the coast of Odisha.
3. MIRV technology allows a single missile to deliver multiple warheads to distinct target locations.

Which of the statements given above are correct?

- (A) 1 and 2 only
- (B) 2 and 3 only
- (C) 1 and 3 only
- (D) 1, 2 and 3

Correct Answer: (D)

Core Memory Fact: Logic:** Mission Divyastra on 11 March 2024 marked India's successful maiden flight test of the indigenously developed Agni-V missile with MIRV capability, placing India among an elite group of nations possessing this strategic... **Ans: D.** 1, 2 and 3

Q2 UPPCS PRELIMS 2024

1. It is a solid-propellant surface-to-surface quasi-ballistic missile for battlefield tactical strike.
2. It has an operational range of 150 to 500 km.
3. In December 2025, India successfully conducted a salvo launch of two Pralay missiles off the Odisha coast.

Which of the statements given above is/are correct?

- (A) 1 and 2 only
- (B) 2 and 3 only
- (C) 1 and 3 only
- (D) 1, 2 and 3

Correct Answer: (D)

Core Memory Fact: Logic:** Pralay is a canisterised, solid-fuel tactical quasi-ballistic missile developed by DRDO. **Ans: D.** 1, 2 and 3

Q3 UP RO/ARO PRELIMS 2023

- (A) It is a joint venture between India's DRDO and Russia's NPO Mashinostroyeniya.
- (B) It operates on a two-stage propulsion system with a solid booster and a liquid ramjet engine.
- (C) In April 2024, India delivered the first export batch of BrahMos missiles to Vietnam.
- (D) It travels at a speed of Mach 2.8 to 3.0.

Correct Answer: (C)

Core Memory Fact: Logic:** Statement (C) is incorrect because the first export customer for BrahMos was the Philippines (under a $375-million contract signed in 2022). **Ans: C.** In April 2024, India delivered the first export batch of BrahMos missiles to Vietnam

Q4 UPPCS PRELIMS 2023

1. It is an informal political understanding among member states to limit the proliferation of missiles and unmanned delivery systems capable of carrying WMDs.
2. It places particular focus on rockets and UAVs capable of delivering a payload of at least 500 kg to a range of at least 300 km.
3. India became a member of the MTCR in 2016.

Which of the statements given above are correct?

- (A) 1 and 2 only
- (B) 2 and 3 only
- (C) 1 and 3 only
- (D) 1, 2 and 3

Correct Answer: (D)

Core Memory Fact: Logic:** MTCR was formed in 1987 by the G-7. **Ans: D.** 1, 2 and 3

Q5 UPPCS PRELIMS 2022

1. Surya Kiran : Nepal
2. Yudh Abhyas : USA
3. Khanjar : Kyrgyzstan
4. Mitra Shakti : Bangladesh

Which of the pairs given above are correctly matched?

- (A) 1, 2 and 3 only
- (B) 2, 3 and 4 only
- (C) 1, 3 and 4 only
- (D) 1, 2, 3 and 4

Correct Answer: (A)

Core Memory Fact: Logic:** Pair 4 is incorrectly matched because Mitra Shakti is conducted between India and Sri Lanka. **Ans: A.** 1, 2 and 3 only

Q6 UP RO/ARO PRELIMS 2021

- (A) Finland
- (B) Sweden
- (C) Ukraine
- (D) North Macedonia

Correct Answer: (B)

Core Memory Fact: Logic:** Standard key matches Sweden. **Ans: B.** Sweden

Q7 UPPCS PRELIMS 2021

- (A) USA and China
- (B) USA and Russia
- (C) Russia and China
- (D) USA and North Korea

Correct Answer: (B)

Core Memory Fact: Logic:** The New START treaty was signed in Prague in April 2010 by President Barack Obama and President Dmitry Medvedev, capping deployed strategic nuclear warheads at 1,550. **Ans: B.** USA and Russia

Q8 UPPCS PRELIMS 2020

- (A) Max Planck
- (B) Niels Bohr
- (C) Albert Einstein
- (D) Louis de Broglie

Correct Answer: (C)
Core Memory Fact: Logic:** Albert Einstein was awarded the 1921 Nobel Prize in Physics specifically for his discovery of the law of the photoelectric effect, not for the theory of relativity. **Ans: C.** Albert Einstein

Q9 UPPCS (MAINS) GS-1

- (A) Sir C.V. Raman
- (B) Meghnad Saha
- (C) Subrahmanyan Chandrasekhar
- (D) Satyendra Nath Bose

Correct Answer: (C)
Core Memory Fact: Logic:** Standard key matches Subrahmanyan Chandrasekhar. **Ans: C.** Subrahmanyan Chandrasekhar

Q10 UPPCS PRELIMS 2019

- (A) Auli, Uttarakhand
- (B) Hanle, Ladakh
- (C) Mount Abu, Rajasthan
- (D) Kodaikanal, Tamil Nadu

Correct Answer: (B)
Core Memory Fact: Logic:** MACE is located at Hanle, Ladakh at an altitude of $4,270\text{ m}$. **Ans: B.** Hanle, Ladakh

Q11 (UPPCS PRE 2018)

The Raman Effect is related to the light rays passing through:
- (A) Only gases
- (B) Only liquids
- (C) Only diamonds
- (D) All transparent media

Correct Answer: (D)
Core Memory Fact: Logic:** (D) All transparent media **Ans: D.** All transparent media

Q12 (UPPCS PRE 2018)

Who among the following Indian scientists invented the 'Crescograph', an instrument to measure the growth in plants?
- (A) P.C. Ray
- (B) J.C. Bose
- (C) C.V. Raman
- (D) Meghnad Saha

Correct Answer: (B)
Core Memory Fact: Logic:** Standard key matches J.C. **Ans: B.** J.C.

Q13 (UPPCS PRE 2018)

National Science Day in India is celebrated on:
- (A) February 28
- (B) March 28
- (C) April 28
- (D) May 28

Correct Answer: (A)
Core Memory Fact: Logic:** (A) February 28 **Ans: A.** February 28

Q14 (UPPCS PRE 2021)

For which of the following scientific discoveries was Albert Einstein awarded the Nobel Prize in Physics?
- (A) Theory of Relativity
- (B) Law of Photoelectric Effect
- (C) Theory of Brownian Motion
- (D) Unified Field Theory

Correct Answer: (B)
Core Memory Fact: Logic:** (B) Law of Photoelectric Effect **Ans: B.** Law of Photoelectric Effect

Q15 (UPPCS PRE 2019)

Who among the following is recognized as the 'Father of the Indian Space Program'?
- (A) Dr. Homi J. Bhabha
- (B) Dr. Vikram Sarabhai
- (C) Dr. A.P.J. Abdul Kalam
- (D) Prof. Satish Dhawan

Correct Answer: (B)
Core Memory Fact: Logic:** Standard key matches Dr. **Ans: B.** Dr.

Chapter 18: Chemistry Topic 1 — Atomic Structure and Periodic Classification — UPPCS & UP RO/ARO Drill

15 Authentic Questions • Direct Verified Answers • Active Recall Memory Locks

🎯 High-Yield PYQ Drill (UPPCS & UP RO/ARO)

Exam Drill Rule: High-yield authentic UPPCS & UP RO/ARO questions. Attempt actively; verify against the prominent Memory Lock box below each question.

Q1 UPPCS PRELIMS 2025

With reference to atomic structure, which of the following statements is/are correct?
1. The neutron was discovered by James Chadwick in 1932.
2. The mass of an electron is approximately $1/1836\text{th}$ of the mass of a proton.
3. The volume of an atomic nucleus occupies over 50% of the total volume of an atom.

Select the correct answer using the code given below:
- (A) 1 and 2 only
- (B) 2 and 3 only
- (C) 1 only
- (D) 1, 2 and 3

Correct Answer: (A)
Core Memory Fact: Logic:** (A) 1 and 2 only **Ans: A.** 1 and 2 only

Q2 UPPCS PRELIMS 2024

Given below are two statements, one labelled as Assertion (A) and the other as Reason (R):
- **Assertion (A):** Chlorine has a higher electron affinity than Fluorine.
- **Reason (R):** Fluorine's compact $2p$ subshell has an exceptionally high electron density, producing strong inter-electronic repulsion against incoming electrons.

Select the correct answer:
- (A) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)
- (C) (A) is true but (R) is false
- (D) (A) is false but (R) is true

Correct Answer: (A)
Core Memory Fact: Logic:** (A) Both (A) and (R) are true and (R) is the correct explanation of (A) **Ans: A.** Both (A) and (R) are true and (R) is the correct explanation of (A)

Q3 **UP RO/ARO PRELIMS 2023**

Match List-I (Subatomic Discovery / Rule) with List-II (Pioneering Scientist):

List-I	List-II
A Law of Octaves	1 Ernest Rutherford
B Modern Periodic Law	2 John Newlands
C Discovery of Atomic Nucleus	3 Henry Moseley
D Uncertainty Principle	4 Werner Heisenberg

Select the correct code:
- (A) A-2, B-3, C-1, D-4
- (B) A-3, B-2, C-1, D-4
- (C) A-2, B-3, C-4, D-1
- (D) A-4, B-3, C-1, D-2

Correct Answer: (A)
Core Memory Fact: Logic:** (A) A-2, B-3, C-1, D-4 **Ans: A.** A-2, B-3, C-1, D-4

Q4 **UPPCS PRELIMS 2023**

Which of the following statements is/are correct regarding periodic trends across a period (left to right)?
1. Atomic radius decreases across a period.
2. Electronegativity increases across a period.
3. Metallic character increases across a period.

Select the correct answer using the code given below:
- (A) 1 and 2 only
- (B) 2 and 3 only
- (C) 1 only
- (D) 1, 2 and 3

Correct Answer: (A)
Core Memory Fact: Logic:** (A) 1 and 2 only **Ans: A.** 1 and 2 only

Q5 **UPPCS PRELIMS 2022**

Which of the following pairs of elements and physical states is **NOT** correctly matched?
- (A) Mercury — Liquid metal
- (B) Bromine — Liquid non-metal
- (C) Iodine — Solid non-metal
- (D) Gallium — Gas at room temperature

Correct Answer: (D)
Core Memory Fact: Logic:** (D) Gallium — Gas at room temperature **Ans: D.** Gallium — Gas at room temperature

Q6 UP RO/ARO PRELIMS 2021

Arrange the following elements in descending order of their abundance by weight in the Earth's crust:
1. Iron
2. Oxygen
3. Silicon
4. Aluminum

Select the correct code:
- (A) $2 > 3 > 4 > 1$
- (B) $3 > 2 > 4 > 1$
- (C) $2 > 4 > 3 > 1$
- (D) $2 > 3 > 1 > 4$

Correct Answer: (A)
Core Memory Fact: Logic:** (A) $2 > 3 > 4 > 1$ **Ans: A.** $2 > 3 > 4 > 1$

Q7 UPPCS PRELIMS 2021

With reference to the periodic table, which group contains the chemically inert Noble Gases?
- (A) Group 1
- (B) Group 16
- (C) Group 17
- (D) Group 18

Correct Answer: (D)
Core Memory Fact: Logic:** (D) Group 18 **Ans: D.** Group 18

Q8 UPPCS PRELIMS 2020

Given below are two statements, one labelled as Assertion (A) and the other as Reason (R):
- **Assertion (A):** The first ionization energy of Nitrogen is higher than that of Oxygen.
- **Reason (R):** A half-filled $2p^3$ subshell possesses extra thermodynamic stability due to symmetrical electron distribution.

Select the correct answer:
- (A) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)
- (C) (A) is true but (R) is false
- (D) (A) is false but (R) is true

Correct Answer: (A)
Core Memory Fact: Logic:** (A) Both (A) and (R) are true and (R) is the correct explanation of (A) **Ans: A.** Both (A) and (R) are true and (R) is the correct explanation of (A)

Q9 UPPCS (MAINS) GS-1

What is the maximum number of electrons that can occupy a d-subshell?
- (A) 2
- (B) 6
- (C) 10
- (D) 14

Correct Answer: (C)
Core Memory Fact: Logic:** Standard key matches 10. **Ans: C.** 10

Q10 UPPCS PRELIMS 2019

Which of the following is the most abundant element by mass in the human body?
- (A) Carbon
- (B) Hydrogen
- (C) Oxygen
- (D) Nitrogen

Correct Answer: (C)
Core Memory Fact: Logic:** Standard key matches Oxygen. **Ans: C.** Oxygen

Q11 UP RO/ARO 2016

Match List-I (Element Type) with List-II (Example Element):

List-I	List-II
A Alkali Metal	1 Silicon
B Halogen	2 Sodium
C Metalloid	3 Chlorine
D Noble Gas	4 Argon

Select the correct code:
- (A) A-2, B-3, C-1, D-4
- (B) A-3, B-2, C-1, D-4
- (C) A-2, B-1, C-3, D-4
- (D) A-4, B-3, C-1, D-2

Correct Answer: (A)
Core Memory Fact: Logic:** (A) A-2, B-3, C-1, D-4 **Ans: A.** A-2, B-3, C-1, D-4

Q12 UPPCS PRELIMS 2018

Which of the following pairs is **NOT** correctly matched?
- (A) Highest ionization energy — Helium
- (B) Highest electronegativity — Fluorine
- (C) Highest electron affinity — Fluorine
- (D) Densest element — Osmium

Correct Answer: (C)
Core Memory Fact: Logic:** (C) Highest electron affinity — Fluorine **Ans: C.** Highest electron affinity — Fluorine

Q13 UPPCS (PRE) 2017

With reference to atomic orbitals, the principal quantum number (n) determines:
- (A) The 3D spatial orientation of the orbital
- (B) The main energy level and size of the orbital
- (C) The intrinsic spin of the electron
- (D) The geometric shape of the orbital

Correct Answer: (B)
Core Memory Fact: Logic:** (B) The main energy level and size of the orbital **Ans: B.** The main energy level and size of the orbital

Q14 UPPCS (PRE) 2016

Given below are two statements, one labelled as Assertion (A) and the other as Reason (R):
- **Assertion (A):** Zirconium (Zr) and Hafnium (Hf) possess nearly identical atomic radii.
- **Reason (R):** Lanthanide contraction compensates for the expected increase in atomic size from the 4d to 5d transition series.

Select the correct answer:
- (A) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)
- (C) (A) is true but (R) is false
- (D) (A) is false but (R) is true

Correct Answer: (A)
Core Memory Fact: Logic:** (A) Both (A) and (R) are true and (R) is the correct explanation of (A) **Ans: A.** Both (A) and (R) are true and (R) is the correct explanation of (A)

Q15 UP RO/ARO 2014

The diagonal relationship in the periodic table is exhibited between which pair of elements?
- (A) Sodium and Potassium
- (B) Lithium and Magnesium
- (C) Boron and Carbon
- (D) Beryllium and Boron

Correct Answer: (B)
Core Memory Fact: Logic:** (B) Lithium and Magnesium **Ans: B.** Lithium and Magnesium

Chapter 19: Chemistry Topic 2 — Matter, Solutions and Purification — UPPCS & UP RO/ARO Drill

15 Authentic Questions • Direct Verified Answers • Active Recall Memory Locks

🎯 High-Yield PYQ Drill (UPPCS & UP RO/ARO)

Exam Drill Rule: High-yield authentic UPPCS & UP RO/ARO questions. Attempt actively; verify against the prominent Memory Lock box below each question.

Q1 UPPCS PRELIMS 2025

What is the fourth state of matter called, and what are its primary constituent particles?
(a) Superfluid; paired neutrons
(b) Plasma; ionized gas containing free electrons and positive ions
(c) Bose-Einstein Condensate; ground state bosons
(d) Liquid crystal; polarized dipoles

Correct Answer: (B)
Core Memory Fact: Logic:** Plasma is an ionized gas created at high temperatures or intense electric fields, consisting of free electrons and positive ions. **Ans: B.** Plasma; ionized gas containing free electrons and positive ions

Q2 UPPCS PRELIMS 2024

Which of the following substances does NOT undergo sublimation under ambient atmospheric conditions?
(a) Camphor
(b) Naphthalene
(c) Sodium chloride
(d) Dry ice (solid CO_2)

Correct Answer: (C)
Core Memory Fact: Logic:** Sodium chloride ($NaCl$) is an ionic solid with high lattice energy (melting point 801°C); it does not sublime under ambient conditions. **Ans: C.** Sodium chloride

Q3 UP RO/ARO PRELIMS 2023

An aqueous solution of copper sulphate is classified as a:
(a) Heterogeneous suspension
(b) Homogeneous true solution
(c) Colloidal sol
(d) Emulsion

Correct Answer: (B)
Core Memory Fact: Logic:** Dissolved copper sulphate dissociates into hydrated Cu^{2+} and $SO4^{2-}$ ions ($<1\text{ nm}$ in diameter). **Ans: B.** Homogeneous true solution

Q4 UPPCS PRELIMS 2023

Why is molality preferred over molarity in reporting high-precision thermodynamic and colligative data?
(a) Molality is easier to calculate in practice
(b) Molality depends strictly on masses, which do not change with temperature
(c) Molality is always numerically smaller than molarity
(d) Molality is only applicable to non-polar solvents

Correct Answer: (B)
Core Memory Fact: Logic:** Molality is defined as moles of solute per kilogram of solvent (mol/kg). **Ans: B.** Molality depends strictly on masses, which do not change with temperature

Q5 UPPCS PRELIMS 2022

Which of the following 0.1 m aqueous solutions will exhibit the lowest freezing point?
(a) 0.1 m Glucose ($C_6H_{12}O_6$)
(b) 0.1 m Sodium chloride ($NaCl$)
(c) 0.1 m Barium chloride ($BaCl_2$)
(d) 0.1 m Aluminium chloride ($AlCl_3$)

Correct Answer: (D)
Core Memory Fact: Logic:** Freezing point depression follows $\Delta Tf = i \cdot Kf \cdot m$. **Ans: D.** 0.1 m Aluminium chloride ($AlCl3$)

Q6 UP RO/ARO PRELIMS 2021

The semi-permeable membrane used in commercial Reverse Osmosis (RO) plants is commonly manufactured from:
(a) Porous ceramic clay
(b) Polyamide or cellulose acetate thin-film composite
(c) Vulcanized rubber
(d) Teflon mesh

Correct Answer: (B)
Core Memory Fact: Logic:** Modern RO membranes are thin-film composite membranes made of cross-linked aromatic polyamides or cellulose acetate. **Ans: B.** Polyamide or cellulose acetate thin-film composite

Q7 UPPCS PRELIMS 2021

Which of the following colloidal systems represents a "Gel"?
(a) Shaving cream
(b) Milk
(c) Cheese
(d) Fog

Correct Answer: (C)
Core Memory Fact: Logic:** Standard key matches Cheese. **Ans: C.** Cheese

Q8 UPPCS PRELIMS 2020

According to the Hardy-Schulze rule, which of the following ions will have the highest coagulating power for a negatively charged arsenious sulphide (As_2S_3) sol?
(a) Na^+
(b) Mg^{2+}
(c) Al^{3+}
(d) K^+

Correct Answer: (C)
Core Memory Fact: Logic:** For a negatively charged sol, coagulation is effected by positive cations. **Ans: C.** Al^{3+}

Q9 UPPCS (MAINS) GS-1

If a beam of light is passed through milk diluted with water, the path of the beam becomes clearly visible. This demonstrates:
(a) Brownian motion
(b) Tyndall effect
(c) Electrophoresis
(d) Dialysis

Correct Answer: (B)
Core Memory Fact: Logic:** Diluted milk contains colloidal fat droplets and protein micelles that scatter visible light photons, illuminating the beam's path. **Ans: B.** Tyndall effect

Q10 UPPCS PRELIMS 2019

The process of separating butter from curd or cream from milk in a domestic setting relies on:
(a) Simple filtration
(b) Centrifugation
(c) Fractional distillation
(d) Sedimentation under normal gravity

Correct Answer: (B)
Core Memory Fact: Logic:** Centrifugation spins the liquid at high angular velocities. **Ans: B.** Centrifugation

Q11 UP RO/ARO 2016

Which of the following is an irreversible chemical change?
(a) Glowing of an electric tungsten filament
(b) Rusting of an iron anchor
(c) Sublimation of iodine crystals
(d) Condensation of steam on a cold mirror

Correct Answer: (B)
Core Memory Fact: Logic:** Rusting of iron forms hydrated ferric oxide ($Fe2O3 \cdot xH2O$) through an electrochemical redox reaction, which cannot be reversed by simple physical cooling or heating. **Ans: B.** Rusting of an iron anchor

Q12 UPPCS PRELIMS 2018

What happens to the boiling point of pure water inside a vacuum chamber when the pressure is reduced to 0.1 atm?
(a) It rises above 100°C
(b) It drops significantly below 100°C
(c) It remains exactly at 100°C
(d) Water freezes and boils simultaneously at 500°C

Correct Answer: (B)
Core Memory Fact: Logic:** Boiling occurs when a liquid's vapor pressure equals external pressure. **Ans: B.** It drops significantly below 100°C

Q13 UPPCS (PRE) 2017

Which of the following mixtures can be cleanly separated using a simple separating funnel?
(a) Ethanol and water
(b) Kerosene and water
(c) Acetone and water
(d) Salt and water

Correct Answer: (A)
Core Memory Fact: Logic:** A separating funnel is used to separate two immiscible liquids that form distinct layers based on density. **Ans: B.** Kerosene and water

Q14 UPPCS (PRE) 2016

The concentration of toxic heavy metals such as arsenic or lead in drinking water is typically measured in:
(a) Molarity (M)
(b) Molality (m)
(c) Parts per million (ppm) or parts per billion (ppb)
(d) Normality (N)

Correct Answer: (C)
Core Memory Fact: Logic:** Trace concentrations of contaminants in water are measured in parts per million ($\text{ppm} = 1\text{ mg/L}$) or parts per billion ($\text{ppb} = 1\text{ }\mu\text{g/L}$). **Ans: C.** Parts per million (ppm) or parts per billion (ppb)

Q15 UP RO/ARO 2014

Which of the following statements about Bose-Einstein Condensate is FALSE?
(a) It was theoretically predicted by Satyendra Nath Bose and Albert Einstein.
(b) It is achieved by heating gases to stellar core temperatures.
(c) It was experimentally realized in 1995 using rubidium and sodium atoms.
(d) In this state, atoms occupy the lowest available quantum ground state.

Correct Answer: (B)
Core Memory Fact: Logic:** Statement (b) is false: BEC is achieved by cooling bosonic gases to near absolute zero ($< 100\text{ nK}$), not by heating them. **Ans: B.** It is achieved by heating gases to stellar core temperatures

Chapter 20: Chemistry Topic 3 — Metals, Non-Metals and Chemical Reactions — UPPCS & UP RO/ARO Drill

15 Authentic Questions • Direct Verified Answers • Active Recall Memory Locks

🎯 High-Yield PYQ Drill (UPPCS & UP RO/ARO)

Exam Drill Rule: High-yield authentic UPPCS & UP RO/ARO questions. Attempt actively; verify against the prominent Memory Lock box below each question.

Q1 UPPCS PRELIMS 2025

Which of the following metals has the lowest melting point and is liquid at room temperature?
(a) Gallium
(b) Caesium
(c) Mercury
(d) Lead

Correct Answer: (C)
Core Memory Fact: Logic:** Standard key matches Mercury. **Ans: C.** Mercury

Q2 UPPCS PRELIMS 2024

Which of the following elements is classified as the densest naturally occurring metal?
(a) Platinum
(b) Osmium
(c) Gold
(d) Uranium

Correct Answer: (B)
Core Memory Fact: Logic:** Standard key matches Osmium. **Ans: B.** Osmium

Q3 UP RO/ARO PRELIMS 2023

Why can sodium and potassium be easily sliced with a common knife?
(a) They have high atomic masses
(b) They have only one valence electron and large atomic radii, leading to weak metallic bonding
(c) They form layered covalent networks like graphite
(d) They contain high concentrations of entrapped hydrogen gas

Correct Answer: (B)
Core Memory Fact: Logic:** Alkali metals possess single valence electrons (ns^1) and large atomic volumes, resulting in low cohesive energy and weak metallic bonding that makes the bulk metal soft. **Ans: B.** They have only one valence electron and large atomic radii, leading to weak metallic bonding

Q4 UPPCS PRELIMS 2023

What is the percentage of elemental Silver (Ag) in the commercial alloy "German Silver"?
(a) 20%
(b) 50%
(c) 0%
(d) 80%

Correct Answer: (C)
Core Memory Fact: Logic:** Standard key matches 0%. **Ans: C.** 0%

Q5 UPPCS PRELIMS 2022

Which of the following non-metals possesses a metallic-like shiny luster?
(a) Sulphur
(b) Carbon (amorphous)
(c) Iodine
(d) Phosphorus

Correct Answer: (C)
Core Memory Fact: Logic:** Standard key matches Iodine. **Ans: C.** Iodine

Q6 UP RO/ARO PRELIMS 2021

The thermite reaction used to weld railway tracks uses which element as a reducing agent?
(a) Carbon coke
(b) Aluminium powder
(c) Magnesium wire
(d) Hydrogen gas

Correct Answer: (B)
Core Memory Fact: Logic:** The reaction is: $Fe2O3 + 2Al \to Al2O3 + 2Fe\text{ (molten)} + \text{heat}$. **Ans: B.** Aluminium powder

Q7 UPPCS PRELIMS 2021

Which of the following metals does NOT evolve hydrogen gas when treated with dilute hydrochloric acid?
(a) Magnesium
(b) Zinc
(c) Iron
(d) Copper

Correct Answer: (D)
Core Memory Fact: Logic:** Standard key matches Copper. **Ans: D.** Copper

Q8 UPPCS PRELIMS 2020

The primary ore of Mercury is:
(a) Bauxite
(b) Galena
(c) Cinnabar
(d) Calamine

Correct Answer: (C)
Core Memory Fact: Logic:** Cinnabar is mercuric sulphide (HgS). **Ans: C.** Cinnabar

Q9 UPPCS (MAINS) GS-1

In the extraction of aluminium by the Hall-Héroult process, cryolite (Na_3AlF_6) is added to alumina to:
(a) Act as an oxidizing agent
(b) Lower the melting point and enhance electrical conductivity
(c) Prevent oxidation of carbon anodes
(d) Form a protective slag layer

Correct Answer: (B)
Core Memory Fact: Logic:** Pure alumina ($Al2O3$) melts at 2050°C and is a poor electrical conductor. **Ans: B.** Lower the melting point and enhance electrical conductivity

Q10 UPPCS PRELIMS 2019

Which of the following alloys is used for soldering electrical connections?
(a) Brass ($Cu + Zn$)
(b) Bronze ($Cu + Sn$)
(c) Solder ($Pb + Sn$)
(d) Nichrome ($Ni + Cr$)

Correct Answer: (C)
Core Memory Fact: Logic:** Solder is an alloy of Lead and Tin ($\sim 50\% : 50\%$) with a low melting point ($\approx 183^\circ\text{C}$), ideal for joining electrical wires. **Ans: C.** Solder ($Pb + Sn$)

Q11 UP RO/ARO 2016

What happens to the mass of an iron nail when it undergoes severe atmospheric rusting?
(a) Decreases
(b) Increases
(c) Remains unchanged
(d) Drops to zero

Correct Answer: (B)
Core Memory Fact: Logic:** Rust is hydrated ferric oxide ($Fe2O3 \cdot xH2O$). **Ans: B.** Increases

Q12 UPPCS PRELIMS 2018

Which element is added to iron to make stainless steel resistant to corrosion?
(a) Sulphur
(b) Lead
(c) Chromium
(d) Tungsten

Correct Answer: (C)
Core Memory Fact: Logic:** Chromium ($\ge 12\%$) forms a microscopic, self-healing passive layer of $Cr2O3$ on the steel surface that blocks oxygen and moisture. **Ans: C.** Chromium

Q13 UPPCS (PRE) 2017

Which of the following pairs of metals will produce an explosive reaction when dropped into cold water?
(a) Copper and Silver
(b) Sodium and Potassium
(c) Iron and Lead
(d) Gold and Platinum

Correct Answer: (B)
Core Memory Fact: Logic:** Sodium and potassium react vigorously and exothermically with cold water, evolving hydrogen gas that catches fire spontaneously. **Ans: B.** Sodium and Potassium

Q14 UPPCS (PRE) 2016

Galvanization provides sacrificial protection to iron because:
(a) Zinc is less electropositive than iron
(b) Zinc has a more negative standard reduction potential than iron
(c) Zinc forms a permanent alloy with iron
(d) Zinc absorbs all atmospheric moisture

Correct Answer: (B)
Core Memory Fact: Logic:** Zinc ($E^\circ = -0.76\text{ V}$) is more easily oxidized than iron ($E^\circ = -0.44\text{ V}$). **Ans: B.** Zinc has a more negative standard reduction potential than iron

Q15 UP RO/ARO 2014

What is the geometry and hybridization of carbon atoms in diamond?
(a) Planar hexagonal, sp^2
(b) Tetrahedral, sp^3
(c) Linear, sp
(d) Octahedral, sp^3d^2

Correct Answer: (B)
Core Memory Fact: Logic:** In diamond, every carbon atom is sp^3 hybridized and forms four strong covalent σ-bonds arranged tetrahedrally, producing a rigid three-dimensional network. **Ans: B.** Tetrahedral, sp^3

Chapter 21: Chemistry Topic 4 — Acids, Bases, Salts and Carbohydrates — UPPCS & UP RO/ARO Drill

15 Authentic Questions • Direct Verified Answers • Active Recall Memory Locks

🎯 High-Yield PYQ Drill (UPPCS & UP RO/ARO)

Exam Drill Rule: High-yield authentic UPPCS & UP RO/ARO questions. Attempt actively; verify against the prominent Memory Lock box below each question.

Q1 UPPCS PRELIMS 2025

What is the mathematical definition of pH?
(a) $\text{pH} = \log_{10}[H^+]$
(b) $\text{pH} = -\log_{10}[H^+]$
(c) $\text{pH} = \ln[H^+]$
(d) $\text{pH} = -\log_{e}[OH^-]$

Correct Answer: (B)
Core Memory Fact: Logic:** $\text{pH} = -\log{10}[H^+] = \log{10}\left(\frac{1}{[H^+]}\right)$, measuring hydronium ion concentration on an inverse logarithmic scale. **Ans: B.** $\text{pH} = -\log{10}[H^+]$

Q2 UPPCS PRELIMS 2024

If the pH of an aqueous solution changes from 4 to 2, the concentration of hydrogen ions $[H^+]$:
(a) Decreases by a factor of 2
(b) Increases by a factor of 2
(c) Increases by 100 times
(d) Decreases by 100 times

Correct Answer: (C)
Core Memory Fact: Logic:** $[H^+] = 10^{-\text{pH}}$. **Ans: C.** Increases by 100 times

Q3 UP RO/ARO PRELIMS 2023

What is the color of phenolphthalein in a dilute aqueous solution of sodium hydroxide?
(a) Colorless
(b) Deep pink / magenta
(c) Yellow
(d) Dark blue

Correct Answer: (B)
Core Memory Fact: Logic:** Phenolphthalein is an acid-base indicator that remains colorless in acidic and neutral solutions (pH $ 10.0$). **Ans: B.** Deep pink / magenta

Q4 UPPCS PRELIMS 2023

Why is tartaric acid mixed with sodium hydrogen carbonate to make commercial baking powder?
(a) To accelerate the release of oxygen gas
(b) To neutralize the alkaline sodium carbonate formed on heating, preventing a bitter taste
(c) To impart an artificial fruit flavor
(d) To act as a preservative against mold

Correct Answer: (B)
Core Memory Fact: Logic:** When baking soda ($NaHCO3$) is heated, it decomposes into $Na2CO3$, $H2O$, and $CO2$. **Ans: B.** To neutralize the alkaline sodium carbonate formed on heating, preventing a bitter taste

Q5 UPPCS PRELIMS 2022

Which of the following substances contains water of crystallization?
(a) Baking Soda ($NaHCO_3$)
(b) Washing Soda ($Na_2CO_3 \cdot 10H_2O$)
(c) Common Salt ($NaCl$)
(d) Slaked Lime ($Ca(OH)_2$)

Correct Answer: (B)
Core Memory Fact: Logic:** Washing soda contains 10 molecules of water of crystallization ($Na2CO3 \cdot 10H2O$). **Ans: B.** Washing Soda ($Na2CO3 \cdot 10H2O$)

Q6 UP RO/ARO PRELIMS 2021

What is formed when gypsum is heated above 200°C (473 K)?
(a) Plaster of Paris
(b) Quicklime
(c) Dead burnt plaster (anhydrous $CaSO_4$)
(d) Slaked lime

Correct Answer: (C)
Core Memory Fact: Logic:** Heating gypsum above 200°C drives off all water of crystallization, leaving anhydrous calcium sulphate ($CaSO4$), commonly termed "dead burnt plaster", which loses its hydraulic setting properties. **Ans: C.** Dead burnt plaster (anhydrous $CaSO4$)

Q7 UPPCS PRELIMS 2021

Which of the following acids is present in stinging nettle leaves and ant bites?
(a) Methanoic acid (formic acid)
(b) Ethanoic acid (acetic acid)
(c) Oxalic acid
(d) Citric acid

Correct Answer: (A)
Core Memory Fact: Logic:** Stinging hairs of nettles and venom sacs of red ants inject methanoic acid ($HCOOH$), causing local inflammation and a burning sensation. **Ans: A.** Methanoic acid (formic acid)

Q8 UPPCS PRELIMS 2020

A solution turns red litmus paper blue. Its pH is most likely:
(a) 1
(b) 4
(c) 5
(d) 10

Correct Answer: (D)
Core Memory Fact: Logic:** Standard key matches 10. **Ans: D.** 10

Q9 UPPCS (MAINS) GS-1

Which of the following compounds is a double salt?
(a) Potassium chloride
(b) Potash alum ($K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$)
(c) Bleaching powder
(d) Copper sulphate

Correct Answer: (B)
Core Memory Fact: Logic:** Potash alum is an equimolar double salt composed of potassium sulphate and aluminium sulphate crystallized with 24 water molecules. **Ans: B.** Potash alum ($K2SO4 \cdot Al2(SO4)3 \cdot 24H2O$)

Q10 UPPCS PRELIMS 2019

The primary acid present in vinegar is:
(a) Acetic acid (CH_3COOH)
(b) Formic acid ($HCOOH$)
(c) Butyric acid
(d) Propionic acid

Correct Answer: (A)
Core Memory Fact: Logic:** Vinegar is an aqueous solution containing $4–8\%$ acetic acid (ethanoic acid). **Ans: A.** Acetic acid ($CH3COOH$)

Q11 UP RO/ARO 2016

Which of the following is classified as a non-reducing sugar?
(a) Maltose
(b) Lactose
(c) Sucrose
(d) Glucose

Correct Answer: (C)
Core Memory Fact: Logic:** Standard key matches Sucrose. **Ans: C.** Sucrose

Q12 UPPCS PRELIMS 2018

What is the artificial sweetener that is approximately 550 times sweeter than cane sugar and excreted unchanged in human urine?
(a) Aspartame
(b) Saccharin
(c) Sucralose
(d) Alitame

Correct Answer: (B)
Core Memory Fact: Logic:** Saccharin (o-sulphobenzimide) was the first synthetic artificial sweetener ($\sim 550\times$ sucrose), passes un-metabolized through the renal system, and provides zero calories. **Ans: B.** Saccharin

Q13 UPPCS (PRE) 2017

Which of the following salts produces an alkaline (basic) solution when dissolved in pure water?
(a) Ammonium chloride (NH_4Cl)
(b) Sodium acetate (CH_3COONa)
(c) Sodium chloride ($NaCl$)
(d) Copper sulphate ($CuSO_4$)

Correct Answer: (B)
Core Memory Fact: Logic:** Sodium acetate is the salt of a weak acid ($CH3COOH$) and a strong base ($NaOH$). **Ans: B.** Sodium acetate ($CH3COONa$)

Q14 UPPCS (PRE) 2016

In the human body, the carbonic acid-bicarbonate buffer system operates to maintain the pH of:
(a) Gastric juice
(b) Blood plasma
(c) Bile juice
(d) Pancreatic juice

Correct Answer: (B)
Core Memory Fact: Logic:** The $H2CO3 / HCO3^-$ buffer system maintains systemic blood plasma pH within the critical $7.35–7.45$ range. **Ans: B.** Blood plasma

Q15 UP RO/ARO 2014

What is the chemical formula of "Green Vitriol"?
(a) $CuSO_4 \cdot 5H_2O$
(b) $FeSO_4 \cdot 7H_2O$
(c) $ZnSO_4 \cdot 7H_2O$
(d) $MgSO_4 \cdot 7H_2O$

Correct Answer: (B)
Core Memory Fact: Logic:** Green Vitriol is ferrous sulphate heptahydrate ($FeSO4 \cdot 7H2O$). **Ans: B.** $FeSO4 \cdot 7H2O$

Chapter 22: Chemistry Topic 5 — Carbon, Organic Compounds, Polymers and Fibres — UPPCS & UP RO/ARO Drill

15 Authentic Questions • Direct Verified Answers • Active Recall Memory Locks

🎯 High-Yield PYQ Drill (UPPCS & UP RO/ARO)

Exam Drill Rule: High-yield authentic UPPCS & UP RO/ARO questions. Attempt actively; verify against the prominent Memory Lock box below each question.

Q1 UPPCS PRELIMS 2025

What is the hybridization of carbon atoms in graphite and diamond, respectively?
(a) sp^3 and sp^2
(b) sp^2 and sp^3
(c) sp and sp^2
(d) sp^3 and sp

Correct Answer: (B)

Core Memory Fact: Logic:** In graphite, each carbon forms 3 coplanar σ-bonds (sp^2, 120°). **Ans: B.** sp^2 and sp^3

Q2 UPPCS PRELIMS 2024

Which of the following allotropes of carbon is thermodynamically the most stable under standard ambient conditions?
(a) Diamond
(b) Fullerene (C_{60})
(c) Graphite
(d) Carbon nanotube

Correct Answer: (C)

Core Memory Fact: Logic:** Graphite is the thermodynamic ground state of elemental carbon at 298.15 K and 1 bar, with standard enthalpy of formation defined as $\Deltaf H^\circ = 0\text{ kJ/mol}$. **Ans: C.** Graphite

Q3 UP RO/ARO PRELIMS 2023

The monomer unit of natural rubber is:
(a) Chloroprene
(b) Isoprene (2-methyl-1,3-butadiene)
(c) Styrene
(d) Ethylene

Correct Answer: (B)

Core Memory Fact: Logic:** Natural rubber is cis-1,4-polyisoprene, formed from the monomer isoprene ($CH2=C(CH3)-CH=CH2$). **Ans: B.** Isoprene (2-methyl-1,3-butadiene)

Q4 UPPCS PRELIMS 2023

What chemical cross-link is formed between polymer chains during the vulcanization of rubber?
(a) Ester bonds
(b) Disulphide bonds ($-S-S-$)
(c) Hydrogen bonds
(d) Peptide bonds

Correct Answer: (B)
Core Memory Fact: Logic:** Heating raw rubber with sulphur creates covalent disulphide bridges across polyisoprene chains, preventing permanent slippage under stress. **Ans: B.** Disulphide bonds ($-S-S-$)

Q5 UPPCS PRELIMS 2022

Which of the following is a semi-synthetic fibre manufactured from plant cellulose?
(a) Nylon-6
(b) Rayon
(c) Polyester
(d) Teflon

Correct Answer: (B)
Core Memory Fact: Logic:** Standard key matches Rayon. **Ans: B.** Rayon

Q6 UP RO/ARO PRELIMS 2021

The polymer "Teflon" is chemically:
(a) Polyvinyl chloride
(b) Polytetrafluoroethylene
(c) Polyacrylonitrile
(d) Polyethylene terephthalate

Correct Answer: (B)
Core Memory Fact: Logic:** Teflon is polytetrafluoroethylene ($[-CF2-CF2-]n$). **Ans: B.** Polytetrafluoroethylene

Q7 UPPCS PRELIMS 2021

Which of the following materials is used for manufacturing heat-resistant handles of frying pans and electric iron bases?
(a) PVC
(b) Polyethylene
(c) Bakelite
(d) Polystyrene

Correct Answer: (C)
Core Memory Fact: Logic:** Bakelite is a thermosetting phenol-formaldehyde resin that does not soften under heat and provides electrical insulation. **Ans: C.** Bakelite

Q8 UPPCS PRELIMS 2020

What is the total number of pentagonal rings present in a single molecule of Buckminsterfullerene (C_{60})?
(a) 20
(b) 12
(c) 60
(d) 10

Correct Answer: (B)
Core Memory Fact: Logic:** Standard key matches 12. **Ans: B.** 12

Q9 UPPCS (MAINS) GS-1

Which of the following polymers is an aromatic polyamide (aramid) used in bulletproof armor?
(a) Terylene
(b) Kevlar
(c) Nylon-6,6
(d) Bakelite

Correct Answer: (B)
Core Memory Fact: Logic:** Standard key matches Kevlar. **Ans: B.** Kevlar

Q10 UPPCS PRELIMS 2019

The Ziegler-Natta catalyst used for low-pressure polymerization of ethylene into HDPE consists of:
(a) $Fe + Mo$
(b) $TiCl_4 + Al(C_2H_5)_3$
(c) V_2O_5
(d) $Pd / BaSO_4$

Correct Answer: (B)
Core Memory Fact: Logic:** The Ziegler-Natta catalyst is a combination of titanium tetrachloride and triethylaluminium ($TiCl4 + Al(C2H5)3$). **Ans: B.** $TiCl4 + Al(C2H5)3$

Q11 UP RO/ARO 2016

Which of the following synthetic rubbers is manufactured from the monomer chloroprene?
(a) Buna-S
(b) Buna-N
(c) Neoprene
(d) Thiokol

Correct Answer: (C)
Core Memory Fact: Logic:** Neoprene is polychloroprene, formed from 2-chloro-1,3-butadiene. **Ans: C.** Neoprene

Q12 UPPCS PRELIMS 2018

What type of plastic cannot be reshaped by heating once it has undergone setting?
(a) Thermoplastic
(b) Thermosetting plastic
(c) Elastomer
(d) Linear polymer

Correct Answer: (B)
Core Memory Fact: Logic:** Thermosetting plastics undergo irreversible chemical cross-linking during setting; subsequent heating causes thermal decomposition without melting. **Ans: B.** Thermosetting plastic

Q13 UPPCS (PRE) 2017

Which of the following is a condensation polymer of ethylene glycol and terephthalic acid?
(a) Nylon-6,6
(b) Terylene (Dacron)
(c) Bakelite
(d) Melamine

Correct Answer: (B)
Core Memory Fact: Logic:** Terylene (polyester) is synthesized by the condensation of ethylene glycol with terephthalic acid. **Ans: B.** Terylene (Dacron)

Q14 UPPCS (PRE) 2016

Natural silk is primarily composed of which protein?
(a) Keratin
(b) Fibroin
(c) Collagen
(d) Hemoglobin

Correct Answer: (B)
Core Memory Fact: Logic:** Standard key matches Fibroin. **Ans: B.** Fibroin

Q15 UP RO/ARO 2014

What is the primary constituent gas of Biogas and Gobar Gas?
(a) Carbon dioxide
(b) Methane
(c) Propane
(d) Butane

Correct Answer: (B)
Core Memory Fact: Logic:** Standard key matches Methane. **Ans: B.** Methane

Chapter 23: Chemistry Topic 6 — Gases, Fuels, Energy and Combustion — UPPCS & UP RO/ARO Drill

15 Authentic Questions • Direct Verified Answers • Active Recall Memory Locks

🎯 High-Yield PYQ Drill (UPPCS & UP RO/ARO)

Exam Drill Rule: High-yield authentic UPPCS & UP RO/ARO questions. Attempt actively; verify against the prominent Memory Lock box below each question.

Q1 UPPCS PRELIMS 2025

What is the primary constituent gas of Liquefied Petroleum Gas (LPG)?
(a) Methane
(b) Butane
(c) Ethylene
(d) Acetylene

Correct Answer: (B)
Core Memory Fact: Logic:** Standard key matches Butane. **Ans: B.** Butane

Q2 UPPCS PRELIMS 2024

Which compound is added to commercial LPG to provide a pungent odor for leak detection?
(a) Methyl alcohol
(b) Ethyl mercaptan (C_2H_5SH)
(c) Acetone
(d) Hydrogen sulphide

Correct Answer: (B)
Core Memory Fact: Logic:** Ethyl mercaptan (ethanethiol) is an organosulfur compound added to odorless LPG as a leak warning agent. **Ans: B.** Ethyl mercaptan ($C2H5SH$)

Q3 UP RO/ARO PRELIMS 2023

What is the primary constituent of Compressed Natural Gas (CNG)?
(a) Methane (CH_4)
(b) Butane (C_4H_{10})
(c) Propane (C_3H_8)
(d) Hydrogen (H_2)

Correct Answer: (A)
Core Memory Fact: Logic:** CNG is compressed natural gas composed of $>85\%$ methane. **Ans: A.** Methane ($CH4$)

Q4 UPPCS PRELIMS 2023

Which fuel possesses the highest calorific value per unit mass?
(a) Methane
(b) Petrol
(c) Hydrogen
(d) LPG

Correct Answer: (C)
Core Memory Fact: Logic:** Hydrogen has a calorific value of $150,000\text{ kJ/kg}$, roughly three times that of petrol ($45,000\text{ kJ/kg}$) or methane ($55,000\text{ kJ/kg}$). **Ans: C.** Hydrogen

Q5 UPPCS PRELIMS 2022

What is the composition of "Water Gas"?
(a) $CO + N_2$
(b) $CO + H_2$
(c) $CO_2 + H_2$
(d) $CH_4 + H_2$

Correct Answer: (B)
Core Memory Fact: Logic:** Water gas (syngas) is an equimolar mixture of carbon monoxide and hydrogen ($CO + H2$) formed by steam passing over red-hot coke. **Ans: B.** $CO + H2$

Q6 UP RO/ARO PRELIMS 2021

Producer gas consists primarily of:
(a) $CO + H_2$
(b) $CO + N_2$
(c) $CO_2 + N_2$
(d) $CH_4 + N_2$

Correct Answer: (B)
Core Memory Fact: Logic:** Producer gas is produced by passing air over red-hot coke, yielding carbon monoxide ($\approx 30\%$) and nitrogen ($\approx 60\%$). **Ans: B.** $CO + N2$

Q7 UPPCS PRELIMS 2021

Which of the following is the highest quality rank of coal with over 90% carbon?
(a) Lignite
(b) Peat
(c) Bituminous
(d) Anthracite

Correct Answer: (D)
Core Memory Fact: Logic:** Anthracite contains $>90–95\%$ carbon, burns with minimal smoke, and has the highest calorific value of all coal types. **Ans: D.** Anthracite

Q8 UPPCS PRELIMS 2020

In which Indian state is the major lignite deposit at Neyveli located?
(a) Jharkhand
(b) Tamil Nadu
(c) Odisha
(d) Madhya Pradesh

Correct Answer: (B)
Core Memory Fact: Logic:** Neyveli is located in Cuddalore district of Tamil Nadu. **Ans: B.** Tamil Nadu

Q9 UPPCS (MAINS) GS-1

Octane number is used to rate the anti-knock quality of:
(a) Diesel
(b) Petrol (Gasoline)
(c) Kerosene
(d) Coal gas

Correct Answer: (B)
Core Memory Fact: Logic:** Octane number measures anti-knock performance of petrol in spark-ignition engines. **Ans: B.** Petrol (Gasoline)

Q10 UPPCS PRELIMS 2019

Cetane number evaluates the ignition quality of which automotive fuel?
(a) Aviation turbine fuel
(b) Diesel
(c) Compressed natural gas
(d) Liquefied petroleum gas

Correct Answer: (B)
Core Memory Fact: Logic:** Standard key matches Diesel. **Ans: B.** Diesel

Q11 UP RO/ARO 2016

What gas is generated by the anaerobic decomposition of cow dung in Gobar gas plants?
(a) Butane
(b) Methane
(c) Acetylene
(d) Carbon monoxide

Correct Answer: (B)
Core Memory Fact: Logic:** Standard key matches Methane. **Ans: B.** Methane

Q12 UPPCS PRELIMS 2018

What is the temperature of the coldest, innermost zone of a candle flame?
(a) Hottest zone of the flame
(b) Lowest temperature zone of the flame
(c) Over 2000°C
(d) Same as the blue zone

Correct Answer: (B)
Core Memory Fact: Logic:** The innermost dark zone contains unburnt wax vapors and lacks oxygen, making it the coolest region of the candle flame. **Ans: B.** Lowest temperature zone of the flame

Q13 UPPCS (PRE) 2017

Why is carbon monoxide classified as a toxic asphyxiant?
(a) It destroys lung alveoli mechanically
(b) It binds to hemoglobin $\approx 200\times$ more strongly than oxygen, forming carboxyhemoglobin
(c) It turns blood into nitric acid
(d) It causes rapid blood clotting

Correct Answer: (B)
Core Memory Fact: Logic:** Carbon monoxide has a high affinity for hemoglobin, forming carboxyhemoglobin and preventing normal oxygen delivery to body tissues. **Ans: B.** It binds to hemoglobin $\approx 200\times$ more strongly than oxygen, forming carboxyhemoglobin

Q14 UPPCS (PRE) 2016

In a fire extinguisher, carbon dioxide extinguishes electrical fires primarily by:
(a) Conducting electrical current safely to ground
(b) Displacing oxygen and blanketing the fuel with a dense, non-combustible gas
(c) Reacting with burning copper to form carbonate
(d) Generating water droplets

Correct Answer: (B)
Core Memory Fact: Logic:** $CO2$ is heavier than air; it blankets the fire, cutting off atmospheric oxygen supply, while expansion cools the fuel. **Ans: B.** Displacing oxygen and blanketing the fuel with a dense, non-combustible gas

Q15 UP RO/ARO 2014

According to Charles's Law, if the absolute temperature of a gas is doubled at constant pressure, its volume will:
(a) Halve
(b) Double
(c) Quadruple
(d) Remain unchanged

Correct Answer: (B)
Core Memory Fact: Logic:** Standard key matches Double. **Ans: B.** Double

Chapter 24: Chemistry Topic 7 — Chemistry in Daily Life, Industry and Agriculture — UPPCS & UP RO/ARO Drill

15 Authentic Questions • Direct Verified Answers • Active Recall Memory Locks

🎯 High-Yield PYQ Drill (UPPCS & UP RO/ARO)

Exam Drill Rule: High-yield authentic UPPCS & UP RO/ARO questions. Attempt actively; verify against the prominent Memory Lock box below each question.

Q1 UPPCS PRELIMS 2025

What is the percentage of nitrogen present in commercial Urea fertilizer?
(a) 21%
(b) 28%
(c) 46%
(d) 60%

Correct Answer: (C)
Core Memory Fact: Logic:** Standard key matches 46%. **Ans: C.** 46%

Q2 UPPCS PRELIMS 2024

Which specialty glass is designed with cerium oxide to cut off harmful ultraviolet radiation?
(a) Soda glass
(b) Flint glass
(c) Crookes glass
(d) Pyrex glass

Correct Answer: (C)
Core Memory Fact: Logic:** Crookes glass contains cerium oxide ($CeO2$), which absorbs UV rays in sunglasses. **Ans: C.** Crookes glass

Q3 UP RO/ARO PRELIMS 2023

Which chemical salt is used as a light-sensitive emulsion on photographic film?
(a) Silver bromide ($AgBr$)
(b) Silver iodide (AgI)
(c) Silver nitrate ($AgNO_3$)
(d) Silver oxide (Ag_2O)

Correct Answer: (A)
Core Memory Fact: Logic:** Silver bromide ($AgBr$) is the light-sensitive compound in photographic emulsions. **Ans: A.** Silver bromide ($AgBr$)

Q4 UPPCS PRELIMS 2023

Which compound is used for cloud seeding to generate artificial rain?
(a) Silver bromide
(b) Silver iodide (AgI)
(c) Sodium chloride only
(d) Calcium carbonate

Correct Answer: (B)
Core Memory Fact: Logic:** Silver iodide (AgI) serves as an ice-nucleating agent in clouds due to its crystalline resemblance to ice. **Ans: B.** Silver iodide (AgI)

Q5 UPPCS PRELIMS 2022

Sodium chlorate ($NaClO_3$) is commercially used in agriculture as a:
(a) Fertilizer
(b) Non-selective herbicide
(c) Rodenticide
(d) Fungicide

Correct Answer: (B)
Core Memory Fact: Logic:** Sodium chlorate is a broad-spectrum herbicide used to clear vegetation. **Ans: B.** Non-selective herbicide

Q6 UP RO/ARO PRELIMS 2021

Zinc phosphide (Zn_3P_2) is widely used as a:
(a) Rodenticide (rat poison)
(b) Fertilizer
(c) Selective herbicide
(d) Fungicide

Correct Answer: (A)
Core Memory Fact: Logic:** Zinc phosphide releases toxic phosphine gas in rodent stomachs, acting as a rodenticide. **Ans: A.** Rodenticide (rat poison)

Q7 UPPCS PRELIMS 2021

Why is DDT banned globally under the Stockholm Convention?
(a) It is too expensive to manufacture
(b) It is non-biodegradable and biomagnifies in food chains
(c) It turns soil alkaline
(d) It causes rust on farm machinery

Correct Answer: (B)
Core Memory Fact: Logic:** DDT persists in the environment and accumulates in food webs, causing ecological damage such as avian eggshell thinning. **Ans: B.** It is non-biodegradable and biomagnifies in food chains

Q8 UPPCS PRELIMS 2020

What is the commercial chemical name of the photographic fixer "Hypo"?
(a) Sodium thiosulphate pentahydrate
(b) Sodium sulphate
(c) Sodium sulphite
(d) Sodium bisulphite

Correct Answer: (A)
Core Memory Fact: Logic:** Hypo is sodium thiosulphate ($Na2S2O3 \cdot 5H2O$), which dissolves unexposed silver bromide from film. **Ans: A.** Sodium thiosulphate pentahydrate

Q9 UPPCS (MAINS) GS-1

Aspirin is chemically:
(a) Acetylsalicylic acid
(b) Methyl salicylate
(c) Phenyl salicylate
(d) Benzoic acid

Correct Answer: (A)
Core Memory Fact: Logic:** Aspirin is acetylsalicylic acid, an analgesic, antipyretic, and antiplatelet drug. **Ans: A.** Acetylsalicylic acid

Q10 UPPCS PRELIMS 2019

What byproduct is generated during the manufacture of soap by saponification?
(a) Methanol
(b) Glycerol (Glycerine)
(c) Ethanol
(d) Acetone

Correct Answer: (B)
Core Memory Fact: Logic:** Saponification of triglycerides yields soap and glycerol. **Ans: B.** Glycerol (Glycerine)

Q11 UP RO/ARO 2016

Which of the following compounds is used as a food preservative in tomato ketchup and fruit squashes?
(a) Sodium benzoate
(b) Sodium chlorate
(c) Calcium carbide
(d) Zinc phosphide

Correct Answer: (A)
Core Memory Fact: Logic:** Sodium benzoate ($C6H5COONa$) inhibits yeast and mold growth in acidic foods. **Ans: A.** Sodium benzoate

Q12 UPPCS PRELIMS 2018

What type of glass contains lead oxide (PbO) and possesses a high refractive index used for lenses and prisms?
(a) Soda glass
(b) Flint glass
(c) Pyrex glass
(d) Crookes glass

Correct Answer: (B)
Core Memory Fact: Logic:** Flint glass contains lead oxide, providing high optical dispersion and refractive index for lenses and prisms. **Ans: B.** Flint glass

Q13 UPPCS (PRE) 2017

Why is gypsum added to Portland cement during the grinding process?
(a) To retard initial setting time
(b) To accelerate hardening
(c) To impart color
(d) To increase water solubility

Correct Answer: (A)
Core Memory Fact: Logic:** Gypsum slows the hydration of tricalcium aluminate, preventing flash setting. **Ans: A.** To retard initial setting time

Q14 UPPCS (PRE) 2016

Which soil amendment is added to reclaim acidic soils ($\text{pH} < 6$)?
(a) Gypsum
(b) Agricultural lime ($CaCO_3 / Ca(OH)_2$)
(c) Urea
(d) Ammonium sulphate

Correct Answer: (B)
Core Memory Fact: Logic:** Liming neutralizes excess H^+ ions in acidic soils. **Ans: B.** Agricultural lime ($CaCO3 / Ca(OH)2$)

Q15 UP RO/ARO 2014

What is added to reclaim alkaline/sodic soils ($\text{pH} > 8.5$)?
(a) Slaked lime
(b) Gypsum ($CaSO_4 \cdot 2H_2O$)
(c) Sodium carbonate
(d) Chalk

Correct Answer: (B)
Core Memory Fact: Logic:** Gypsum supplies calcium ions to displace sodium from sodic soils, restoring soil structure. **Ans: B.** Gypsum ($CaSO4 \cdot 2H2O$)

Chapter 25: Chemistry Topic 8 — Nuclear Chemistry, Explosives and Environment — UPPCS & UP RO/ARO Drill

15 Authentic Questions • Direct Verified Answers • Active Recall Memory Locks

🎯 High-Yield PYQ Drill (UPPCS & UP RO/ARO)

Exam Drill Rule: High-yield authentic UPPCS & UP RO/ARO questions. Attempt actively; verify against the prominent Memory Lock box below each question.

Q1 UPPCS PRELIMS 2025

Radioactivity was discovered in 1896 by:
(a) Marie Curie
(b) Henri Becquerel
(c) Ernest Rutherford
(d) Wilhelm Roentgen

Correct Answer: (B)

Core Memory Fact: Logic:** Henri Becquerel discovered radioactivity using uranium salts; Marie and Pierre Curie later isolated polonium and radium. **Ans: B.** Henri Becquerel

Q2 UPPCS PRELIMS 2024

Which nuclear emission consists of doubly charged helium nuclei ($^4_2He^{2+}$)?
(a) Beta particles
(b) Alpha particles
(c) Gamma rays
(d) Neutrons

Correct Answer: (B)

Core Memory Fact: Logic:** Alpha particles are helium-4 nuclei ($2\text{ protons} + 2\text{ neutrons}$) with a $+2$ charge. **Ans: B.** Alpha particles

Q3 UP RO/ARO PRELIMS 2023

The half-life of Carbon-14 used in radiocarbon dating is approximately:
(a) $1,000\text{ years}$
(b) $5,730\text{ years}$
(c) $100,000\text{ years}$
(d) 4.5 billion years

Correct Answer: (B)

Core Memory Fact: Logic:** ^{14}C has a half-life of $\approx 5,730\text{ years}$, suitable for dating organic artifacts up to $\approx 50,000\text{ years}$ old. **Ans: B.** $5,730\text{ years}$

Q4 UPPCS PRELIMS 2023

What is the SI unit of radioactivity?
(a) Curie
(b) Rutherford
(c) Becquerel
(d) Sievert

Correct Answer: (C)
Core Memory Fact: Logic:** The Becquerel ($\text{Bq} = 1\text{ disintegration/sec}$) is the SI unit. **Ans: C.** Becquerel

Q5 UPPCS PRELIMS 2022

Nuclear fission involves:
(a) Joining light nuclei into heavier ones
(b) Splitting of a heavy unstable nucleus into smaller fragments with energy release
(c) Chemical reaction with oxygen
(d) Radioactive decay of electrons

Correct Answer: (B)
Core Memory Fact: Logic:** Fission splits heavy nuclei (^{235}U) into lighter fragments upon neutron absorption, releasing energy. **Ans: B.** Splitting of a heavy unstable nucleus into smaller fragments with energy release

Q6 UP RO/ARO PRELIMS 2021

The primary energy principle governing the Hydrogen bomb is:
(a) Nuclear Fission
(b) Nuclear Fusion
(c) Chemical Combustion
(d) Alpha radiation

Correct Answer: (B)
Core Memory Fact: Logic:** Hydrogen bombs operate on thermonuclear fusion, initiated by an internal fission trigger. **Ans: B.** Nuclear Fusion

Q7 UPPCS PRELIMS 2021

Which of the following substances was used by Alfred Nobel to stabilize nitroglycerin into Dynamite?
(a) Kieselguhr (diatomaceous earth)
(b) Sulphur dust
(c) Cement
(d) Activated charcoal

Correct Answer: (A)
Core Memory Fact: Logic:** Liquid nitroglycerin was absorbed into porous kieselguhr to form shock-stable dynamite. **Ans: A.** Kieselguhr (diatomaceous earth)

Q8 UPPCS PRELIMS 2020

The chemical explosive RDX is chemically known as:
(a) Trinitrotoluene
(b) Cyclotrimethylenetrinitramine
(c) Pentaerythritol tetranitrate
(d) Nitroglycerin

Correct Answer: (B)
Core Memory Fact: Logic:** RDX is cyclotrimethylenetrinitramine ($C3H6N6O6$), also known as Hexogen or T4. **Ans: B.** Cyclotrimethylenetrinitramine

Q9 UPPCS (MAINS) GS-1

Acid rain is defined as precipitation with a pH below:
(a) 7.0
(b) 6.5
(c) 5.6
(d) 4.0

Correct Answer: (C)
Core Memory Fact: Logic:** Standard key matches 5.6. **Ans: C.** 5.6

Q10 UPPCS PRELIMS 2019

The two primary atmospheric pollutant gases responsible for acid rain are:
(a) Carbon monoxide and Methane
(b) Sulphur dioxide (SO_2) and Nitrogen oxides (NO_x)
(c) Ozone and Argon
(d) Chlorine and Fluorine

Correct Answer: (B)
Core Memory Fact: Logic:** $SO2$ and NOx dissolve in atmospheric moisture to form $H2SO4$ and $HNO3$, driving acid precipitation. **Ans: B.** Sulphur dioxide ($SO2$) and Nitrogen oxides (NOx)

Q11 UP RO/ARO 2016

Photochemical smog is an oxidizing smog that contains which of the following secondary pollutants?
(a) Ozone and Peroxyacetyl Nitrate (PAN)
(b) Sulphur dioxide and Coal soot
(c) Methane and Argon
(d) Lead oxide

Correct Answer: (A)
Core Memory Fact: Logic:** Photochemical smog forms under sunlight from NOx and hydrocarbons, generating ozone and PAN. **Ans: A.** Ozone and Peroxyacetyl Nitrate (PAN)

Q12 UPPCS PRELIMS 2018

In which atmospheric layer is the protective ozone layer concentrated?
(a) Troposphere
(b) Stratosphere
(c) Mesosphere
(d) Thermosphere

Correct Answer: (B)
Core Memory Fact: Logic:** The ozone layer is situated in the stratosphere ($15–35\text{ km}$ altitude). **Ans: B.** Stratosphere

Q13 UPPCS (PRE) 2017

Which reactive chemical species released by CFCs catalytically destroys ozone molecules?
(a) Free chlorine radicals ($Cl^\bullet$)
(b) Sodium ions
(c) Nitrogen molecules
(d) Argon atoms

Correct Answer: (A)
Core Memory Fact: Logic:** UV photolysis of CFCs releases free chlorine radicals ($Cl^\bullet$) that catalytically destroy thousands of ozone molecules. **Ans: A.** Free chlorine radicals ($Cl^\bullet$)

Q14 UPPCS (PRE) 2016

Which of the following is NOT a greenhouse gas?
(a) Methane (CH_4)
(b) Carbon dioxide (CO_2)
(c) Water vapor (H_2O)
(d) Nitrogen gas (N_2)

Correct Answer: (D)
Core Memory Fact: Logic:** Homonuclear diatomic nitrogen ($N2$) lacks an infrared-active dipole moment and does not absorb thermal radiation. **Ans: D.** Nitrogen gas ($N2$)

Q15 UP RO/ARO 2014

The environmental tragedy "Minamata Disease" was caused by toxic bioaccumulation of:
(a) Methylmercury
(b) Cadmium
(c) Arsenic
(d) Lead

Correct Answer: (A)
Core Memory Fact: Logic:** Methylmercury bioaccumulated through marine food webs, causing neurological poisoning in Minamata Bay. **Ans: A.** Methylmercury

Chapter 26: Chemistry Topic 9 — Discoveries, Scientists and Miscellaneous Chemistry — UPPCS & UP RO/ARO Drill

15 Authentic Questions • Direct Verified Answers • Active Recall Memory Locks

🎯 High-Yield PYQ Drill (UPPCS & UP RO/ARO)

Exam Drill Rule: High-yield authentic UPPCS & UP RO/ARO questions. Attempt actively; verify against the prominent Memory Lock box below each question.

Q1 UPPCS PRELIMS 2025

Who is celebrated as the "Father of Modern Chemistry"?
(a) John Dalton
(b) Antoine Lavoisier
(c) Robert Boyle
(d) Jöns Jacob Berzelius

Correct Answer: (B)

Core Memory Fact: Logic:** Antoine Lavoisier formulated the law of conservation of mass, named oxygen and hydrogen, and published the first modern list of elements. **Ans: B.** Antoine Lavoisier

Q2 UPPCS PRELIMS 2024

Which chemical salt is used as a light-sensitive indelible marker in Indian election voting ink?
(a) Silver chloride
(b) Silver nitrate (Lunar Caustic)
(c) Silver iodide
(d) Silver bromide

Correct Answer: (B)

Core Memory Fact: Logic:** Silver nitrate ($AgNO3$) photoreduces on skin into metallic silver, leaving an indelible stain. **Ans: B.** Silver nitrate (Lunar Caustic)

Q3 UP RO/ARO PRELIMS 2023

What is the primary function of Heavy Water (D_2O) in Pressurized Heavy Water Reactors (PHWRs)?
(a) Nuclear fuel
(b) Moderator to slow down neutrons
(c) Absorber to stop neutrons
(d) Radiation detector

Correct Answer: (B)

Core Memory Fact: Logic:** Heavy water slows fast fission neutrons to thermal speeds without absorbing them, sustaining the chain reaction. **Ans: B.** Moderator to slow down neutrons

Q4 UPPCS PRELIMS 2023

What is the chemical formula of "Sindoor" (Red Lead / Minium)?
(a) PbO
(b) PbO_2
(c) Pb_3O_4
(d) $Pb(NO_3)_2$

Correct Answer: (C)
Core Memory Fact: Logic:** Standard key matches $Pb3O4$. **Ans: C.** $Pb3O4$

Q5 UPPCS PRELIMS 2022

Temporary hardness of water is caused by dissolved:
(a) Chlorides of calcium and magnesium
(b) Bicarbonates of calcium and magnesium
(c) Sulphates of calcium and magnesium
(d) Nitrates of sodium

Correct Answer: (B)
Core Memory Fact: Logic:** Bicarbonates ($Ca(HCO3)2, Mg(HCO3)2$) cause temporary hardness, removable by boiling. **Ans: B.** Bicarbonates of calcium and magnesium

Q6 UP RO/ARO PRELIMS 2021

Which reagent is added in Clark's process to remove temporary water hardness?
(a) Common salt ($NaCl$)
(b) Slaked lime ($Ca(OH)_2$)
(c) Hydrochloric acid
(d) Nitric acid

Correct Answer: (B)
Core Memory Fact: Logic:** Clark's process uses slaked lime to precipitate calcium carbonate. **Ans: B.** Slaked lime ($Ca(OH)2$)

Q7 UPPCS PRELIMS 2021

Which of the following compounds is commonly known as "Calomel"?
(a) $HgCl_2$
(b) Hg_2Cl_2
(c) HgO
(d) HgS

Correct Answer: (B)
Core Memory Fact: Logic:** Calomel is mercurous chloride ($Hg2Cl2$). **Ans: B.** $Hg2Cl2$

Q8 UPPCS PRELIMS 2020

"Dry Ice" is the solid form of which gas?
(a) Nitrogen
(b) Carbon dioxide
(c) Oxygen
(d) Sulphur dioxide

Correct Answer: (B)
Core Memory Fact: Logic:** Dry ice is solid $CO2$, subliming at -78.5°C. **Ans: B.** Carbon dioxide

Q9 UPPCS (MAINS) GS-1

Who synthesized the first organic compound (urea) from an inorganic substance in 1828?
(a) Friedrich Wöhler
(b) Jöns Jacob Berzelius
(c) August Kekulé
(d) Louis Pasteur

Correct Answer: (A)
Core Memory Fact: Logic:** Friedrich Wöhler synthesized urea from ammonium cyanate, overturning the Vital Force Theory. **Ans: A.** Friedrich Wöhler

Q10 UPPCS PRELIMS 2019

Acharya Prafulla Chandra Ray is famous for discovering and synthesizing:
(a) Heavy water
(b) Mercurous nitrite
(c) Benzene
(d) Dynamite

Correct Answer: (B)
Core Memory Fact: Logic:** Standard key matches Mercurous nitrite. **Ans: B.** Mercurous nitrite

Q11 UP RO/ARO 2016

What is the red dye used as a coloring agent in commercial red ink?
(a) Methylene blue
(b) Eosin
(c) Malachite green
(d) Indigo

Correct Answer: (B)
Core Memory Fact: Logic:** Standard key matches Eosin. **Ans: B.** Eosin

Q12 UPPCS PRELIMS 2018

What is the common name for Lead Monoxide (PbO)?
(a) Sindoor
(b) Litharge
(c) Calomel
(d) Corrosive sublimate

Correct Answer: (B)
Core Memory Fact: Logic:** Litharge is lead monoxide (PbO). **Ans: B.** Litharge

Q13 UPPCS (PRE) 2017

Which radioactive isotope of hydrogen has a half-life of $\approx 12.3\text{ years}$?
(a) Protium
(b) Deuterium
(c) Tritium
(d) Helium-3

Correct Answer: (C)
Core Memory Fact: Logic:** Standard key matches Tritium. **Ans: C.** Tritium

Q14 UPPCS (PRE) 2016

Calgon used in water softening is chemically:
(a) Sodium hexametaphosphate
(b) Sodium carbonate
(c) Sodium chloride
(d) Calcium carbonate

Correct Answer: (A)
Core Memory Fact: Logic:** Calgon is sodium hexametaphosphate ($Na6P6O{18}$), which sequesters calcium and magnesium ions. **Ans: A.** Sodium hexametaphosphate

Q15 UP RO/ARO 2014

The exhausted zeolite bed in water softening is regenerated using an aqueous solution of:
(a) Common salt ($NaCl$ brine)
(b) Sulphuric acid
(c) Sodium hydroxide
(d) Bleaching powder

Correct Answer: (A)
Core Memory Fact: Logic:** Zeolite beds are regenerated by backwashing with a 10% $NaCl$ brine solution to replace captured Ca^{2+} with Na^+. **Ans: A.** Common salt ($NaCl$ brine)